

Week 1:
Intro to Network Routing
&
Static Routing



- Networks have had a **significant** impact on our lives

Roman Roads



Roman Wagon



Format – Physical layout of the wagon compartments

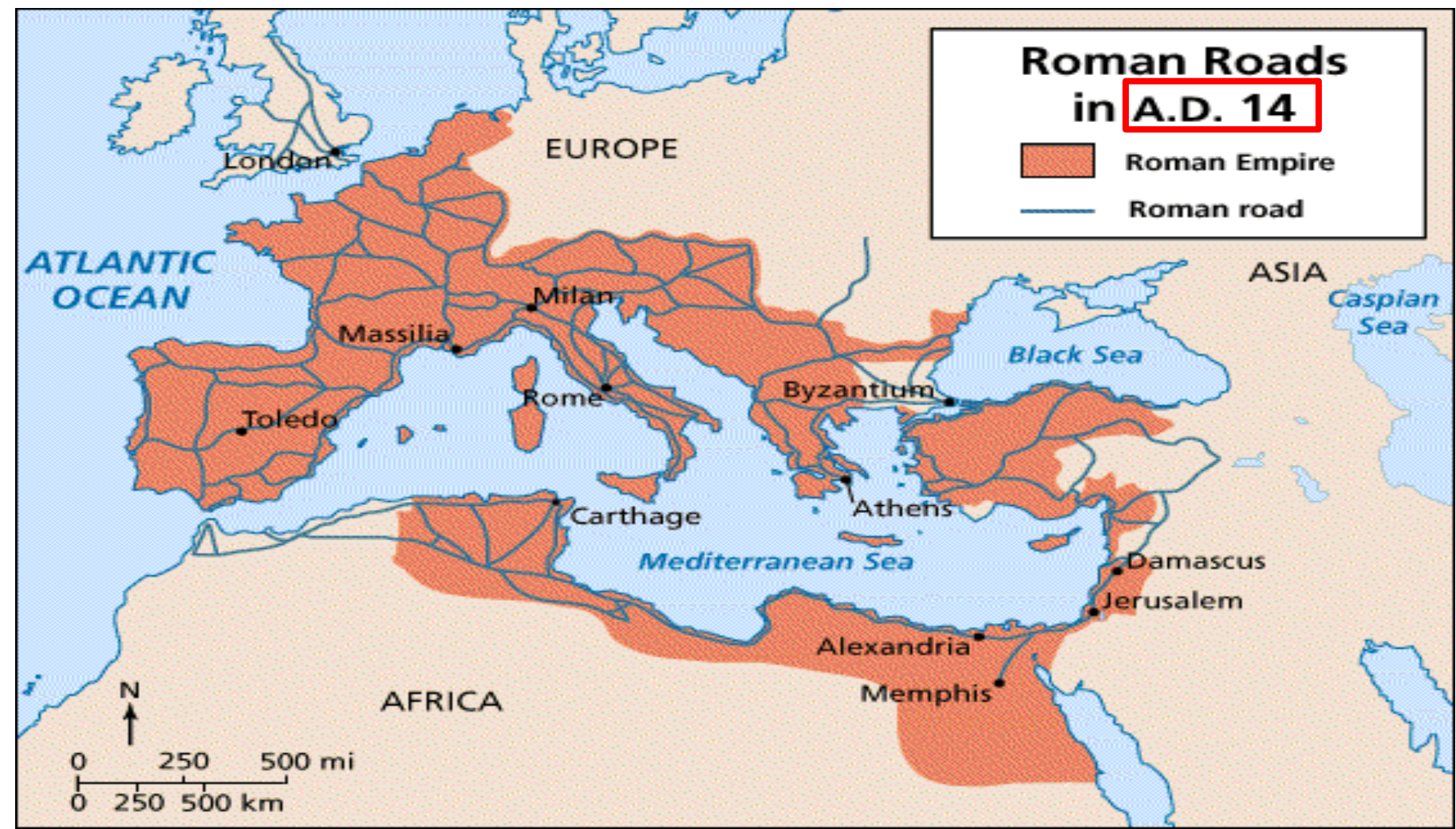
Payload – farm produce, buildings materials, people

Interface – Roads, Canals, Rails



The Roman Road Network

- Reduced the TIME to communicate



Router – The Sign Post – Static Routing

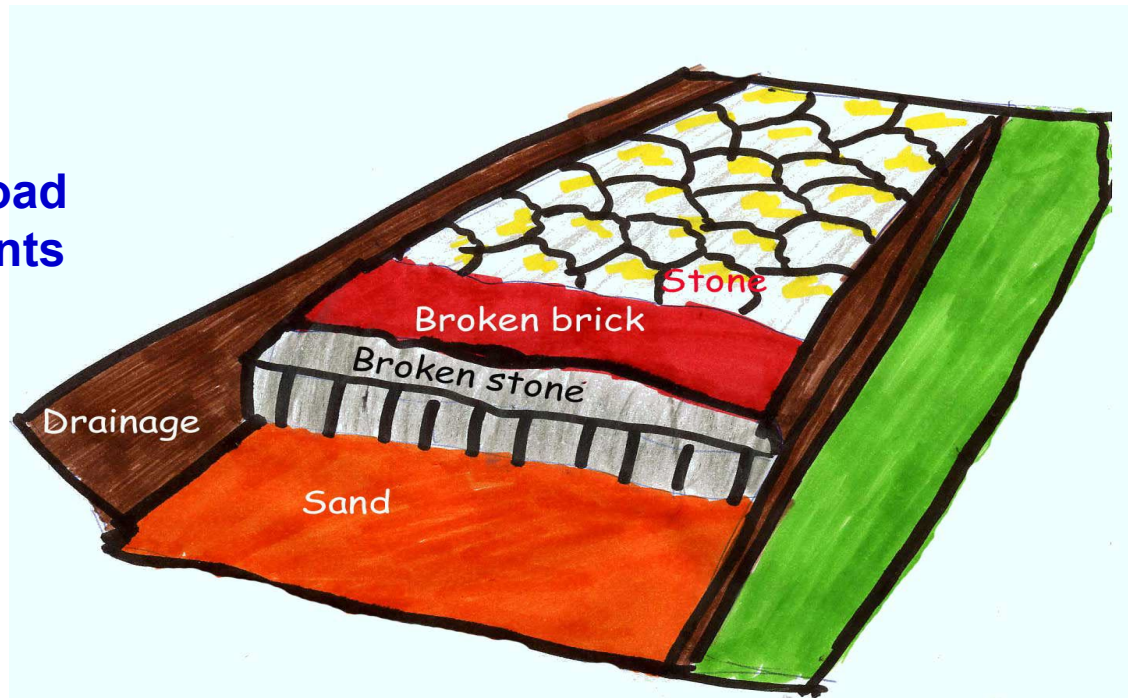


Terms referred to when discussing Networks



- **Topology** – the arrangement and interconnection of roads
- **Speed** – rate of travel, km/h (bits/sec), along a road
- **Cost** – cost of components, installation and maintenance

Roman Road Components



Terms referred to when discussing Networks



- **Security**
 - How to protect the network
- **Availability**
 - A measure of probability the network is available for use
 - Flood makes road impassable
- **Scalability**
 - How easy it is to accommodate more users
 - Increase the width of road
- **Reliability**
 - A measure of the MTBF

- Web servers
- IP Phones
- Mobile Phones
- Video conferencing
- Interactive gaming
- Online shopping
- Online coursework

From Physical to Virtual Transport Networks

Physical – Deliver Mug via Road Network

44 Burwood Rd
Hawthorn, Vic

Sydney
Warehouse



Virtual - Send Instructions to make Mug
via Internet

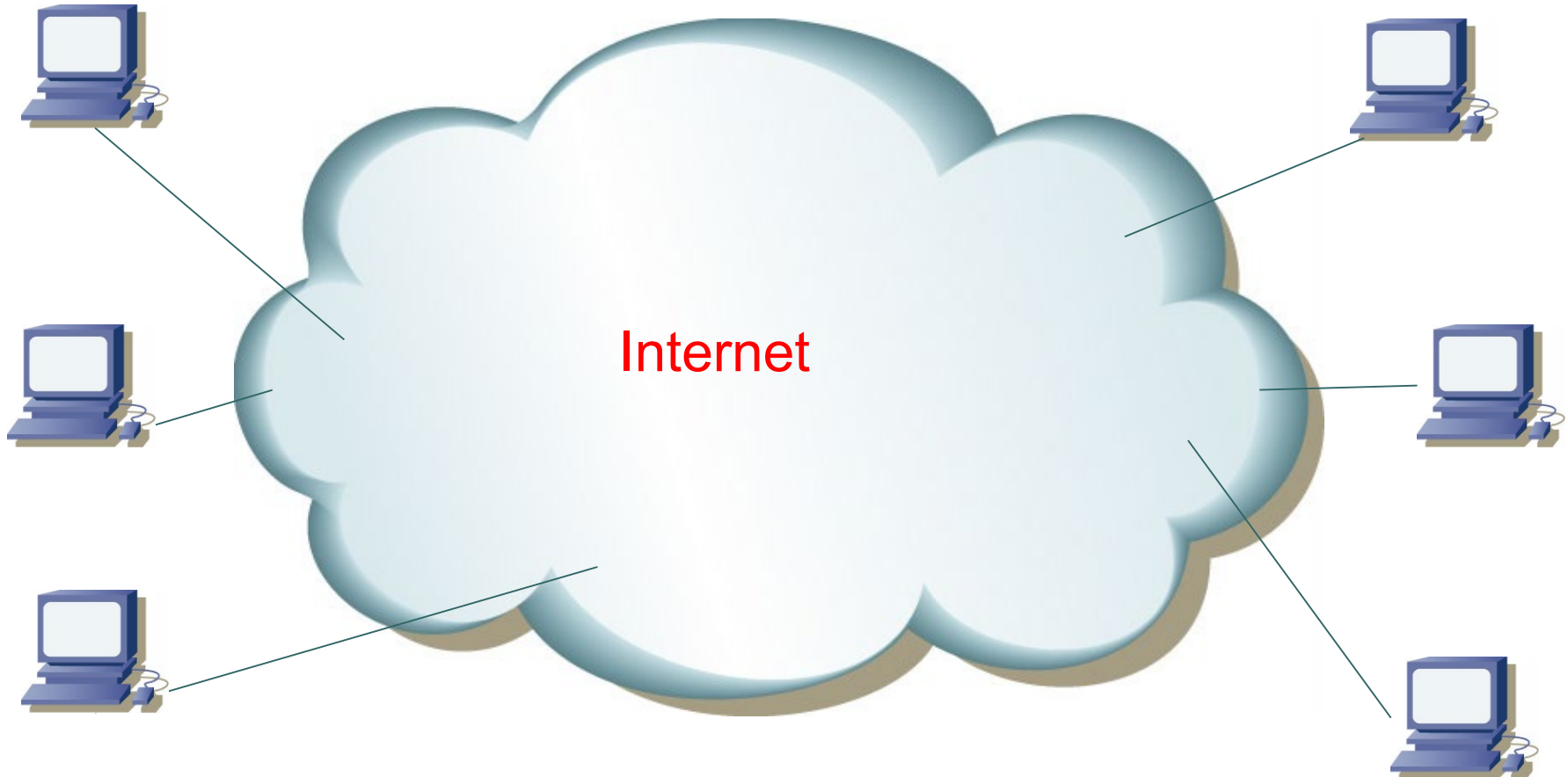
Sydney
Database





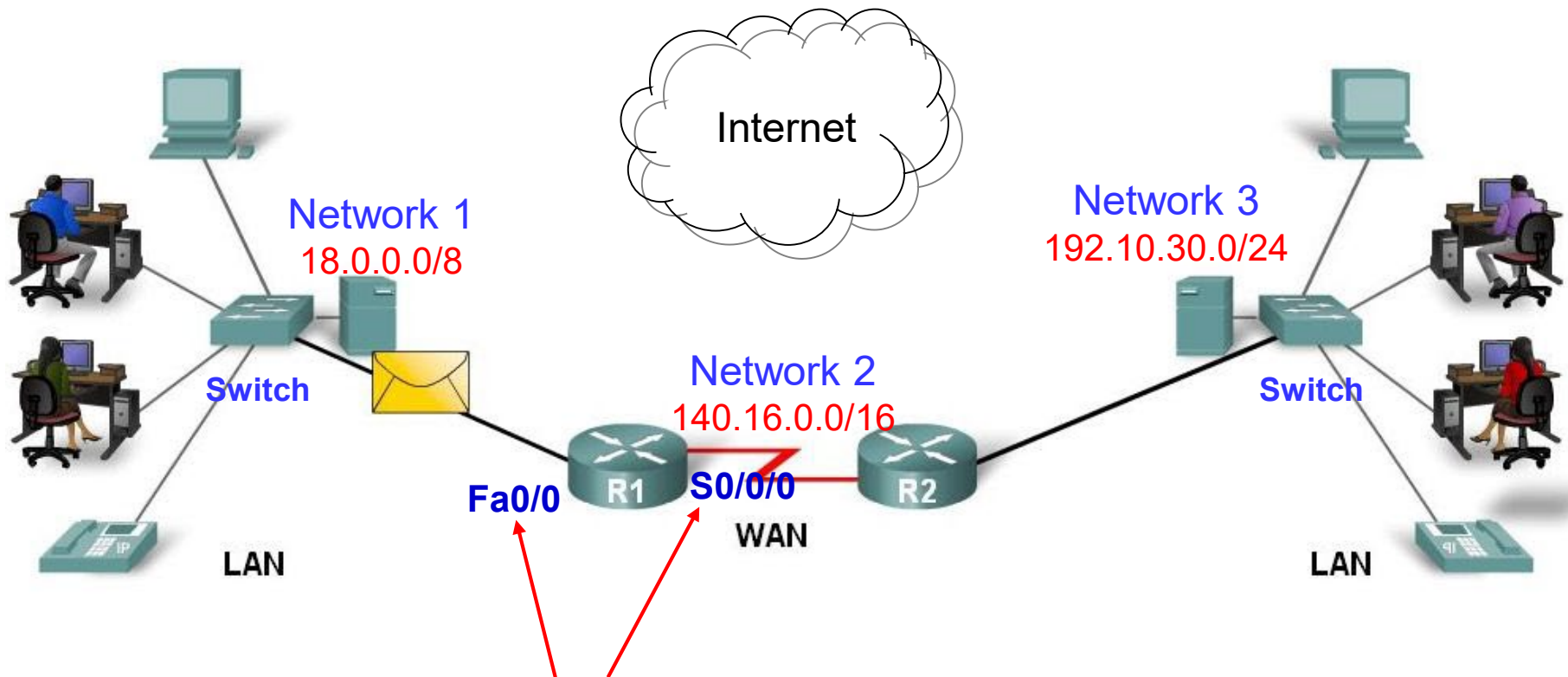
The
Router
is at the
core
of our
Virtual Transport Network

- The **Internet** is too large to have all hosts together as one (large) network



Routers

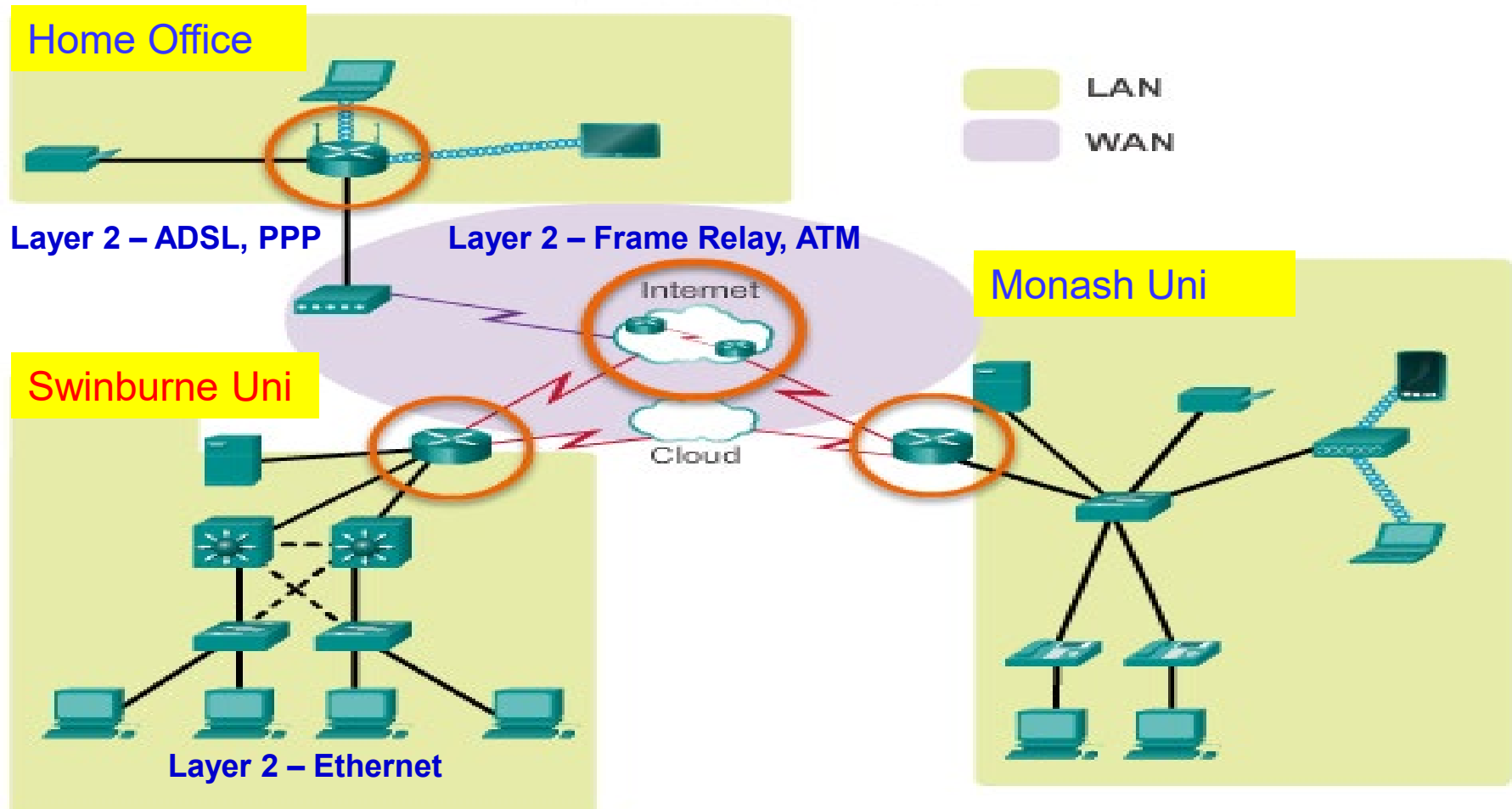
- The **Internet** consists of **many** networks.
- Routers are devices, that route **IP** packets **between** these networks



A Router Interface is a GATEWAY – Entry/Exit into/out of a (sub)network

Routers Interconnect Networks

- Routers can connect multiple networks/subnets.
- Routers have multiple interfaces, **each on a different IP (sub)network.**



Routers are Computers



Routers are **specialized** computers containing the following required components to operate:

- Central processing unit (CPU)
- Operating system (OS) - Routers use **Cisco IOS**
- Memory and storage
 - **RAM** – Random Access – **Running IOS, Routing Tables**
 - **ROM** – Read Only Memory – **Boot up Instructions**
 - **NVRAM** – Non-Volatile Random Access Memory – **Startup configuration**
 - **Flash** – Non-Volatile – **IOS file**
 - **Hard Disk ?**

Routers are Computers



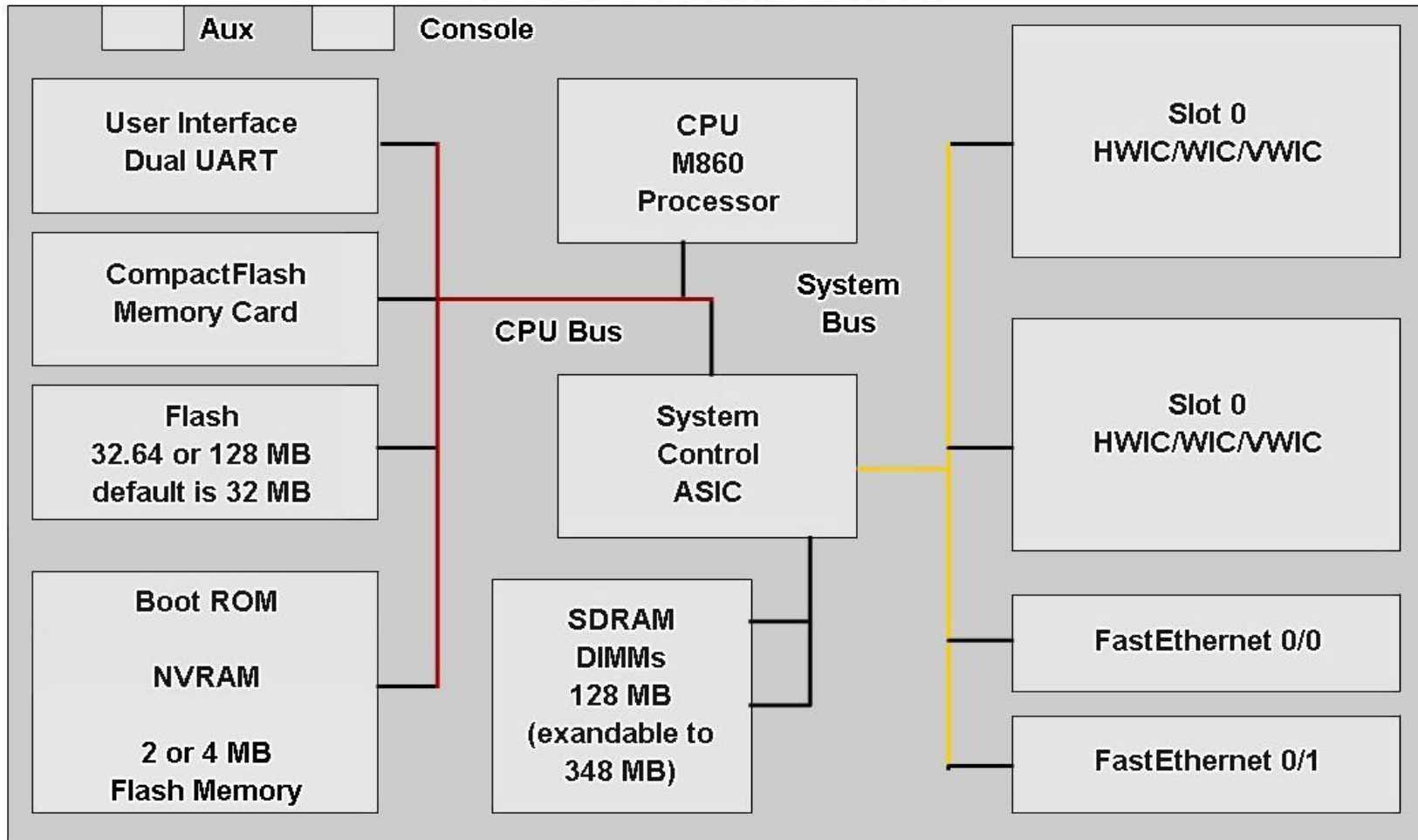
Memory	Volatile / Non-Volatile	Stores
RAM	Volatile	• Running IOS • Running configuration file • IP routing and ARP tables • Packet buffer
ROM	Non-Volatile	• Bootup instructions • Basic diagnostic software • Limited IOS
NVRAM	Non-Volatile	• Startup configuration file
Flash	Non-Volatile	• IOS • Other system files

Routers are Computers



Running **special OS** to perform a specific task – **Cisco IOS**

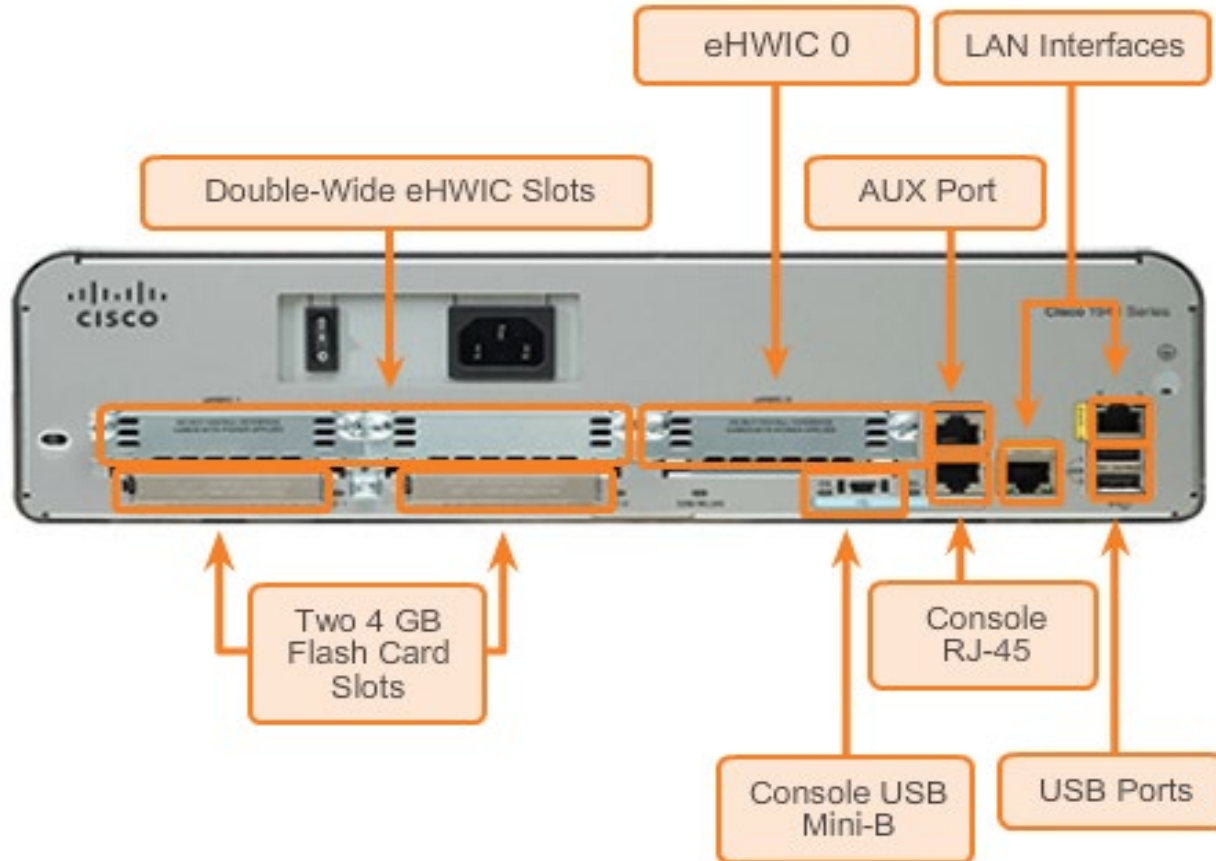
Hardware components of a router



Routers are Computers

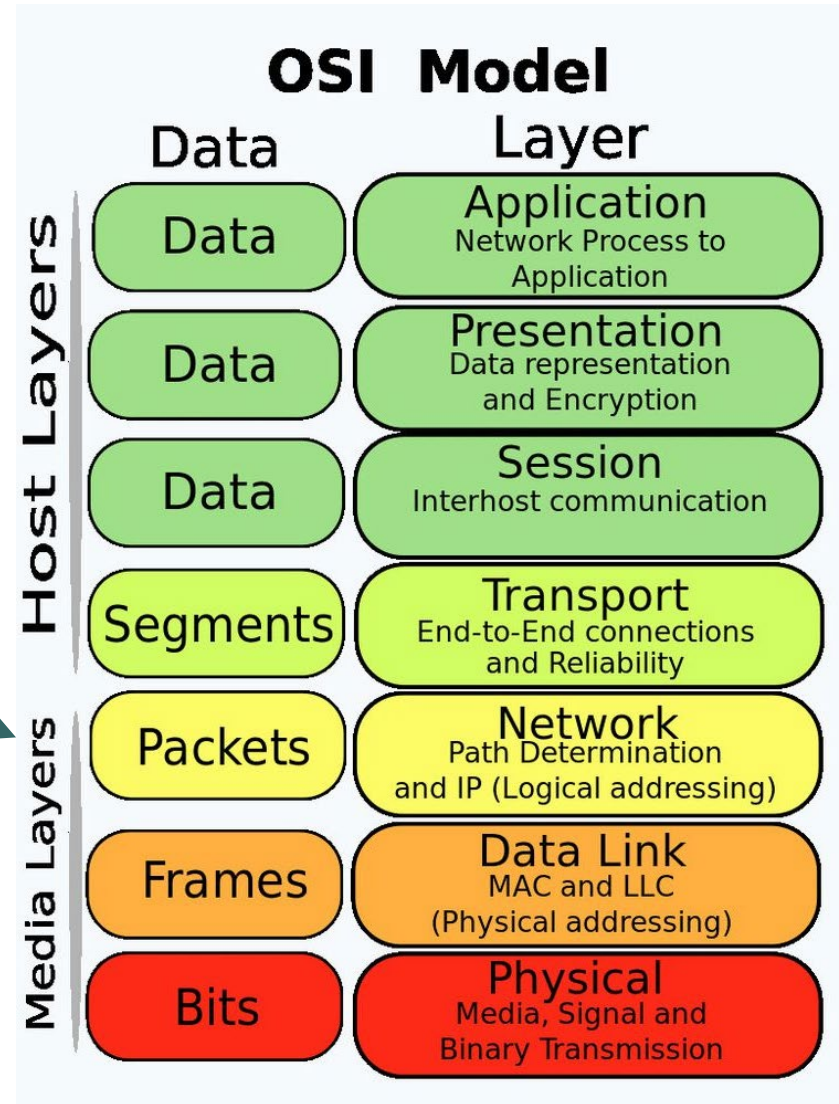
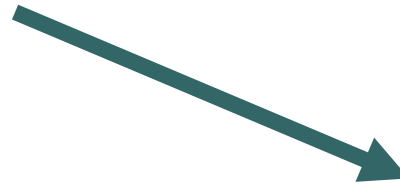
Routers use **specialized ports** and **network interface cards** to interconnect to other networks.

Back Panel of a Router





The Routing Process





The Router – Primary Task

- The Routing process operates at Layer 3
 - Routing IP packets
 - Responsible for routing IP packets to the next hop – eventually to the intended destination
 - Dropping IP packets
 - drop a packet when Time To Live (TTL) in IPv4 packet expires – deduct 1 from TTL when packet enters router
 - this ensures that if a IP packet is caught in a routing loop
 - or has been forwarded through many routers and has not reached its destination
 - it is dropped hence reducing congestion in the network

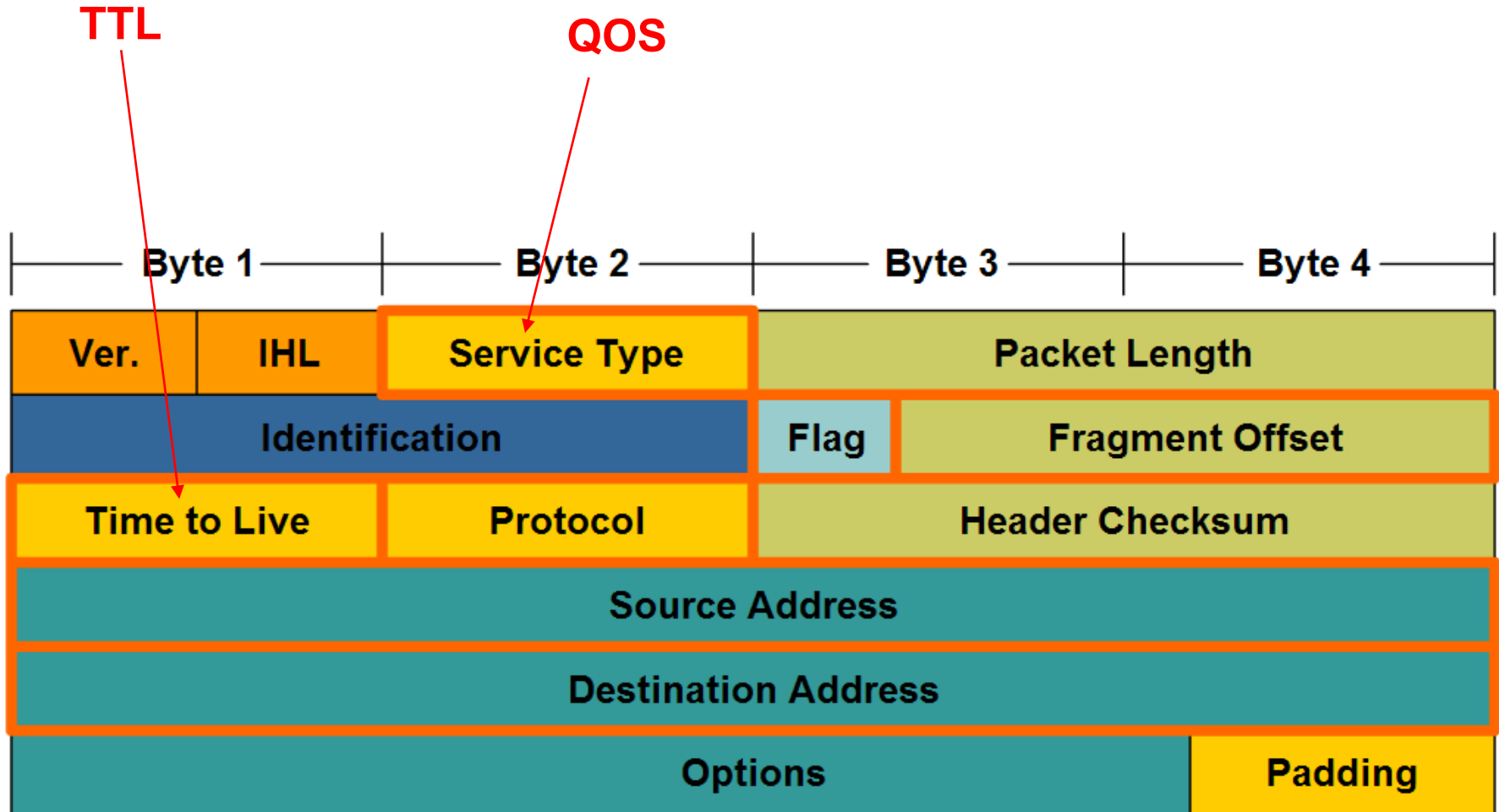


- Security
 - Manage basic **firewalling** of IP packets via **Access Control Lists (ACLs)**
 - Quality of Service
 - Provide **QOS** to some IP packets
 - IP Packets carry **data**, **voice**, **video**
- via **Service Type** field in IP Header

IP Header Fields for QOS and TTL



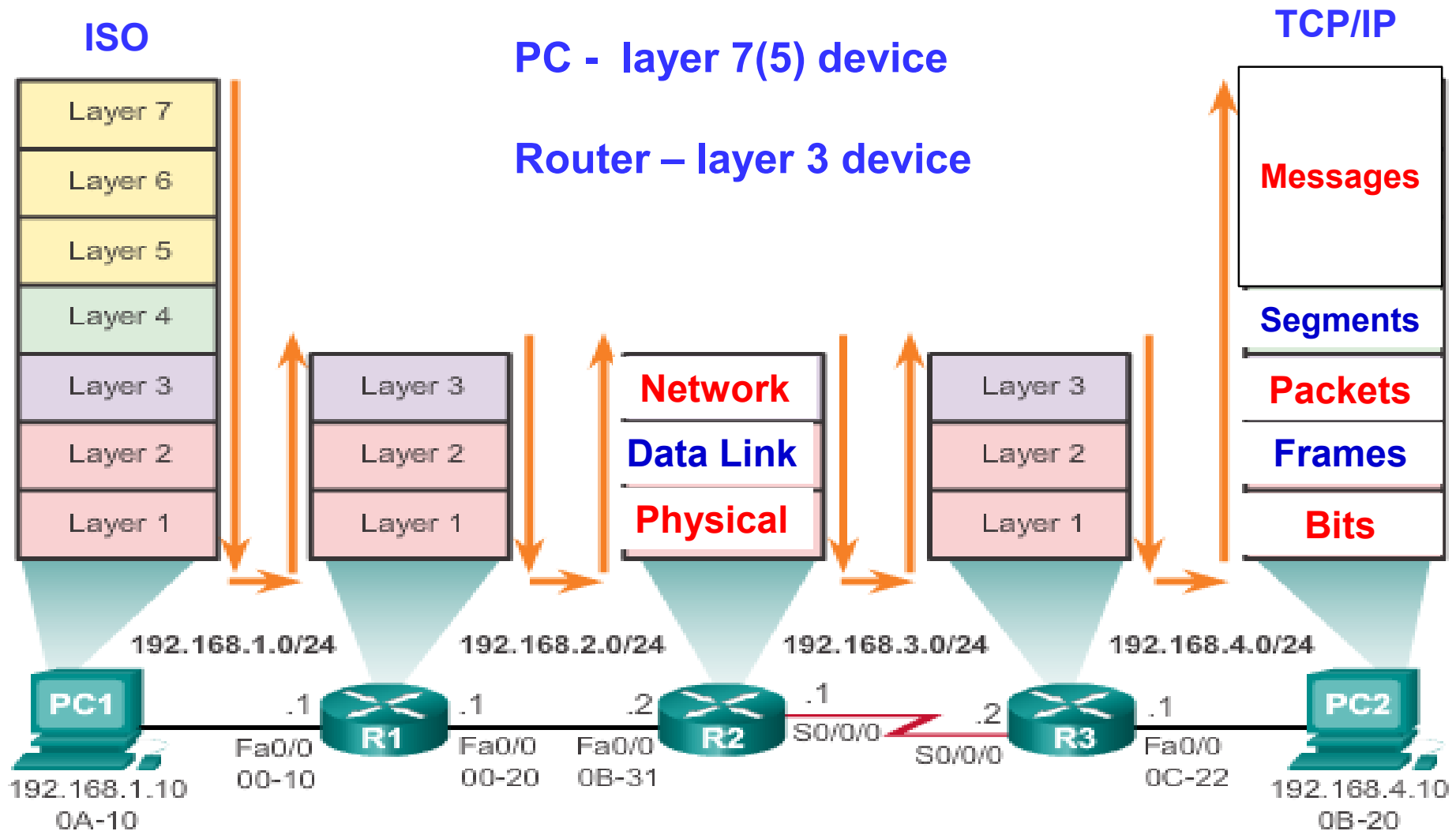
IPv4 Packet Header Fields





Routing

Encapsulating and De-Encapsulating Packets





Based on the **information** in the **Routing Table**, the Router

1. Makes a **Routing** Decision

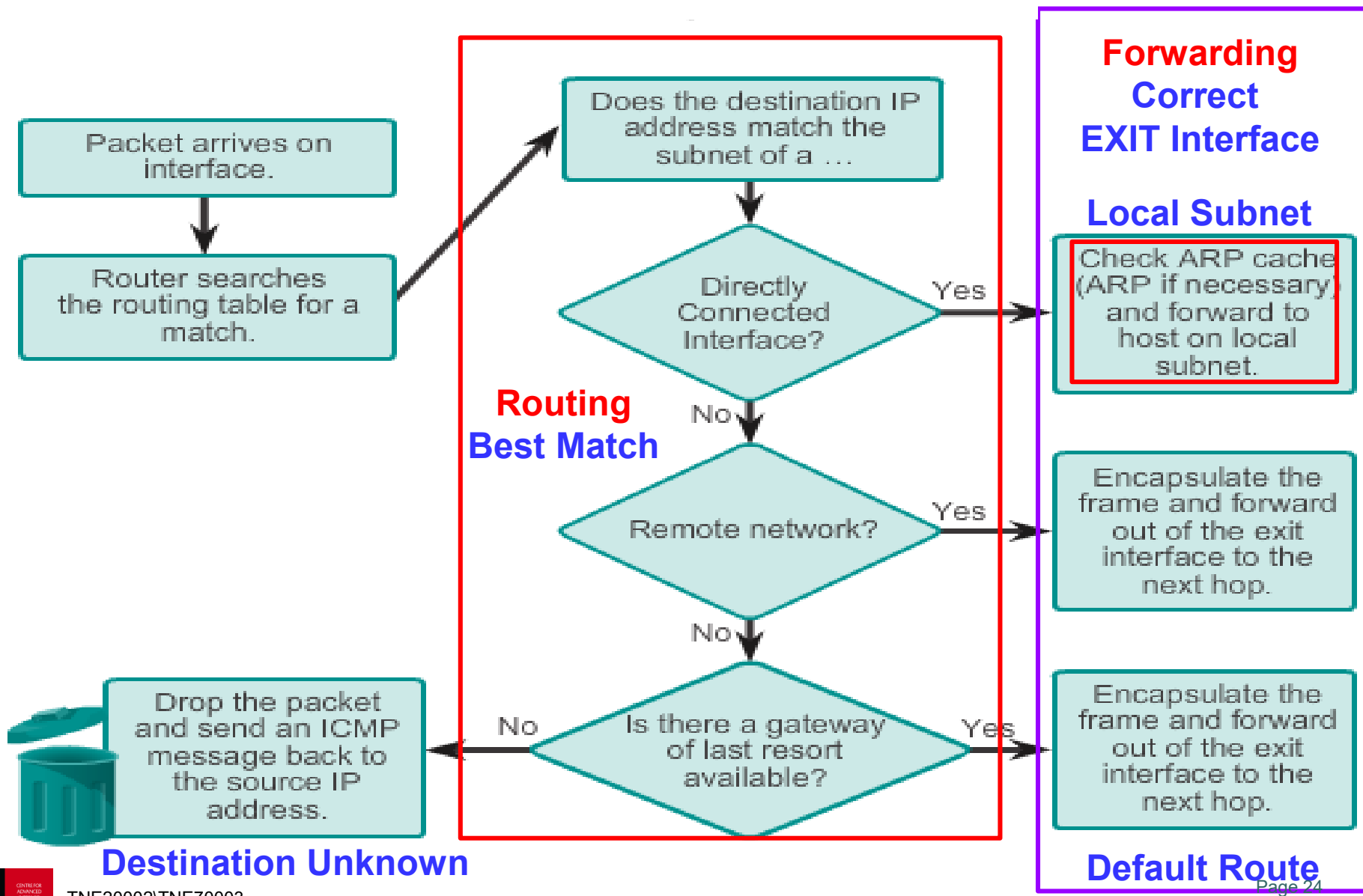
- Determines the **Best Match**, between the **destination network address** of a **packet** and a **network address entry** **IN** the routing table

2. Makes a **Forwarding** Decision

- Determines the **correct exit interface** then **forwards** packets to their **destination** network.



The Router - Decisions

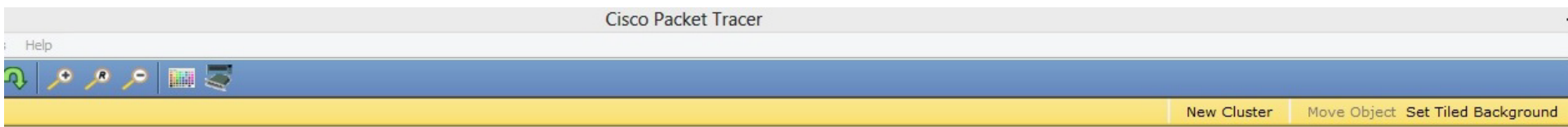


Router ARP Table – Local Subnets

Forwarding a packet through FastEthernet Interfaces



- Display ARP Table on Router - **show arp**



Router0

Physical Config CLI

IOS Command Line Interface

```
Router(config-if)#int f1/1
Router(config-if)#int f1/1
%Invalid interface type and number
Router(config)#int f1/0
Router(config-if)#ip add
Router(config-if)#ip address 100.25.44.1 255.255.255.0
Router(config-if)#no shut

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet1/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up

Router(config-if)#
Router(config-if)#
Router(config-if)#
Router(config-if)#do show arp
```

Protocol	Address	Age (min)	Hardware Addr	Type	Interface
Internet	100.25.34.1	-	0010.112B.5E2C	ARPA	FastEthernet0/0
Internet	100.25.34.25	0	0060.7026.A3C6	ARPA	FastEthernet0/0
Internet	100.25.44.1	-	000A.41DC.7757	ARPA	FastEthernet1/0
Internet	100.25.44.10	0	0090.0CE3.EBB2	ARPA	FastEthernet1/0

Router(config-if)#

Copy Paste

Two Local Subnets
100.25.34.0/24
100.25.44.0/24

PC ARP Table



- Display ARP Table on PC in Command Window - **arp -a**

```
Windows PowerShell
Copyright (C) 2013 Microsoft Corporation. All rights reserved.

PS C:\Users\Peter> arp -a

Interface: 10.0.0.20 --- 0x4
Internet Address      Physical Address      Type
10.0.0.3              e8-40-f2-31-e1-a5    dynamic
10.0.0.6              00-1b-a9-f2-e7-67    dynamic
10.0.0.137            0a-76-ff-4e-d9-b5    dynamic
10.0.0.138            08-76-ff-4e-d9-b5    dynamic
10.0.0.255            ff-ff-ff-ff-ff-ff    static
224.0.0.22            01-00-5e-00-00-16    static
224.0.0.251           01-00-5e-00-00-fb    static
224.0.0.252           01-00-5e-00-00-fc    static
239.255.255.250       01-00-5e-7f-ff-fa    static
239.255.255.253       01-00-5e-7f-ff-fd    static
255.255.255.255       ff-ff-ff-ff-ff-ff    static

PS C:\Users\Peter>
```




Based on the **information** in the **Routing Table**, the Router

1. Makes a **Routing** Decision

- Determines the **Best Match**, between the **destination network address** of a packet and a **network address entry** in the routing table, then determines whether:
 - Directly connected interface ? → Host on Local Subnet
 - Remote network ?
 - No Match ? -> Is there a gateway of last resort ?

2. Makes a **Forwarding** Decision

- Determines the **correct exit interface** then **forwards** packets to their destination network.

The Router - Best Match



- The **Best Match** is the one that has the **most** number of left most bits matching between the **destination IP address** and the **routing entry** in the routing table.

Best Match is the Preferred Route

IP Packet Destination		172.16.0.10	10101100.00010000.00000000.00001010
12 bits	Route 1	172.16.0.0/12	10101100.00010000.00000000.00000000
18 bits	Route 2	172.16.0.0/18	10101100.00010000.00000000.00000000
26 bits	Route 3	172.16.0.0/26	10101100.00010000.00000000.00000000

Best Match to IP Packet Destination

26 bits Match

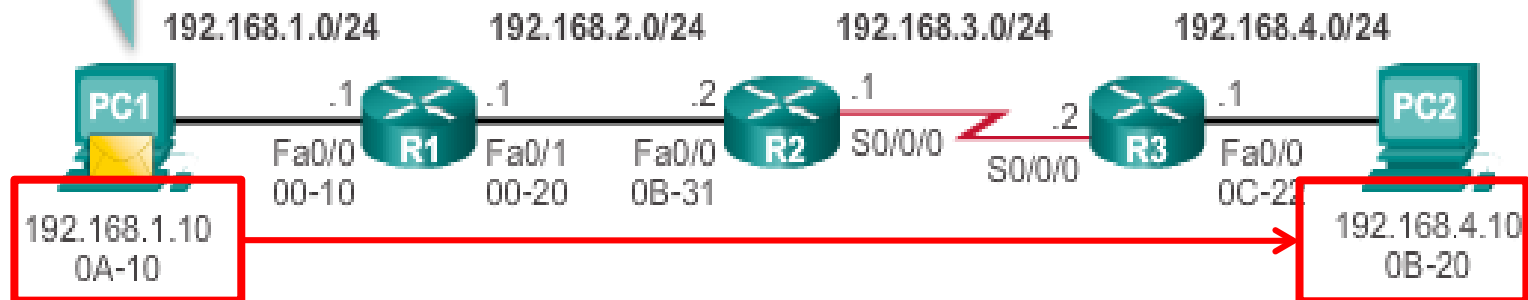




Sending a Packet

PC1 Sends a Packet to PC2

Because PC2 is on different network, I will encapsulate the packet and send it to the router on MY network. Let me find that MAC address....



Layer 2 Data Link Frame

Dest. MAC 00-10	Source MAC 0A-10	Type 800	Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	Trailer
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PC1's ARP Cache for R1

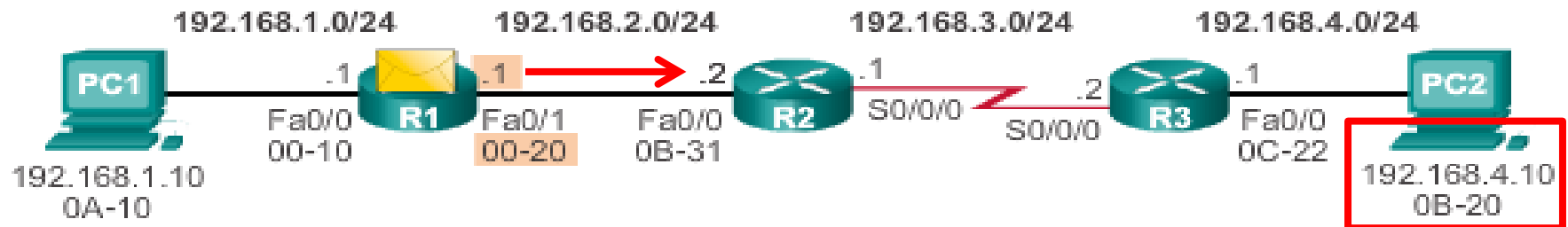
IP Address	MAC Address
192.168.1.1	00-10

Gateway router IP address

Forward to the Next Hop



R1 Forwards the Packet to **R2**



Layer 2 Data Link Frame

Packet's Layer 3 data

Dest. MAC 0B-31	Source MAC 00-20	Type 800	Source IP 192.168.1.10	Dest. IP 192.168.4.10	IP fields	Data	Trailer
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R1's Routing Table

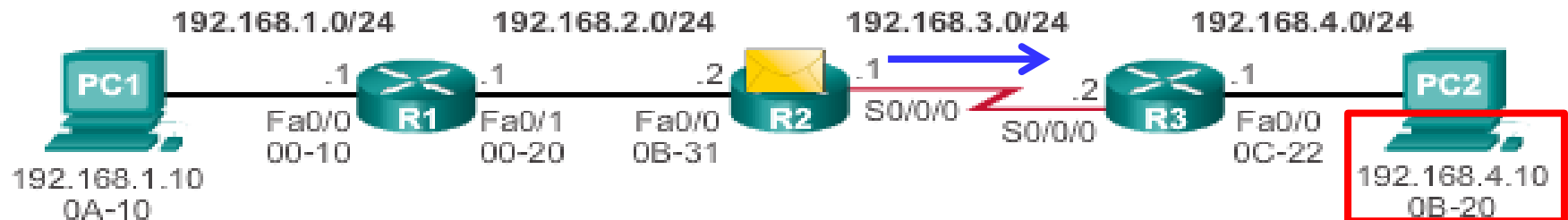
Network	Hops	Next-hop-IP	Exit Interface
192.168.1.0/24	0	Dir. Connect.	Fa0/0
192.168.2.0/24	0	Dir. Connect.	Fa0/1
192.168.3.0/24	1	192.168.2.2	Fa0/1
192.168.4.0/24	2	192.168.2.2	Fa0/1

Match



Packet Routing

R2 Forwards the Packet to **R3**



Layer 2 Data Link Frame

Packet's Layer 3 data

Address	Control	Type	Source IP	Dest. IP	IP fields	Data	Trailer
0x8F	0x00	800	192.168.1.10	192.168.4.10			

R2's Routing Table

Network	Hops	Next-hop-IP	Exit Interface
192.168.1.0/24	1	192.168.3.1	Fa0/0/0
192.168.2.0/24	0	Dir. Connect.	Fa0/0/0
192.168.3.0/24	0	Dir. Connect.	S0/0/0
192.168.4.0/24	1	192.162.3.2	S0/0/0

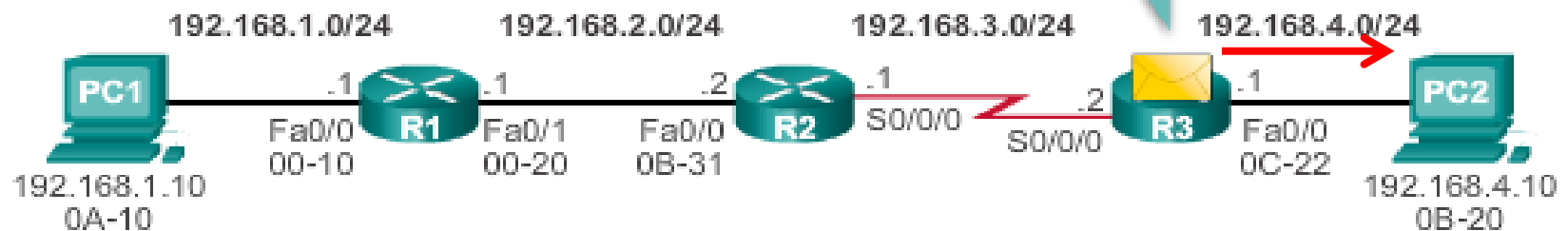
Match

Reaching the Destination



R3 Forwards the Packet to PC2

My ARP table tells me that PC2 uses MAC address 0B-20.



Layer 2 Data Link Frame

Dest. MAC	Source MAC	Type	Source IP	Dest. IP	IP fields	Data	Trailer
0B-20	0C-22	800	192.168.1.10	192.168.4.10			

Packet's Layer 3 data

R3's ARP Cache	
IP Address	MAC Address
192.168.4.10	0B-20

R3's Routing Table			
Network	Hops	Next-hop-IP	Exit Interface
192.168.1.0/24v	2	192.168.3.1	S0/0/0
192.168.2.0/24	1	192.162.3.1	S0/0/0
192.168.3.0/24	0	Dir. Connect.	S0/0/0
192.168.4.0/24	0	Dir. Connect.	Fa0/0

Match



The Routing Protocol



Path Determination - Best Path

- The **Best path** to a given network is determined by a **routing protocol** calculating the **cost**.
- The **cost calculation** can involve a number of **metrics**.
- The **Best path** to a network is the **path** with the **lowest cost**.



- **Hop Count**: counts the number of routers a (IP) packet must traverse
- **Bandwidth**: Influences path selection by **preferring** the path with the **highest** bandwidth
- **Load**: Considers the **traffic utilization** of a link
- **Delay**: Considers the **time** a packet takes to **traverse** a path
- **Reliability**: Assesses the **probability of** a link **failure**



The
COST
is
calculated
based on the
metrics
used by the
Routing
protocol



Routing Information Protocol

RIP

cost

based on metric

Hop count



Open Shortest Path First

OSPF

cost

based on metric

cumulative bandwidth

from source to destination



Enhanced Interior Gateway Routing Protocol

EIGRP

cost

based on metrics

Bandwidth, delay, load, reliability



Routing Table - Best Paths

Only the
Best paths
to given destination networks,
as determined by a routing protocol,
are placed in
the routing table



Path Determination - Load Balancing

When a router has **two or more paths** to a destination with **equal cost**, then the router forwards the packets using both paths equally:

- Equal cost load balancing can improve network performance.
- Equal cost load balancing can be configured to use both dynamic routing protocols and static routes.
- **RIP, OSPF** and **EIGRP** support equal cost load balancing.

Path Determination - Administrative Distance



If **multiple routing protocols** are configured on a router, the **routing entry** installed in the routing table **is the one** with the **lowest Administrative Distance (AD)**:

- A static route with an **AD of 1** is **more reliable** than an EIGRP-discovered route with an **AD of 90**.
- A directly connected route with an **AD of 0** is **more reliable** than a static route with an **AD of 1**.

Route Source	Administrative Distance
Connected	0
Static	1
EIGRP summary route	5
External BGP	20
Internal EIGRP	90
OSPF	110
RIP	120
IS-IS	115
External EIGRP	170
Internal BGP	200

Default Administrative Distances



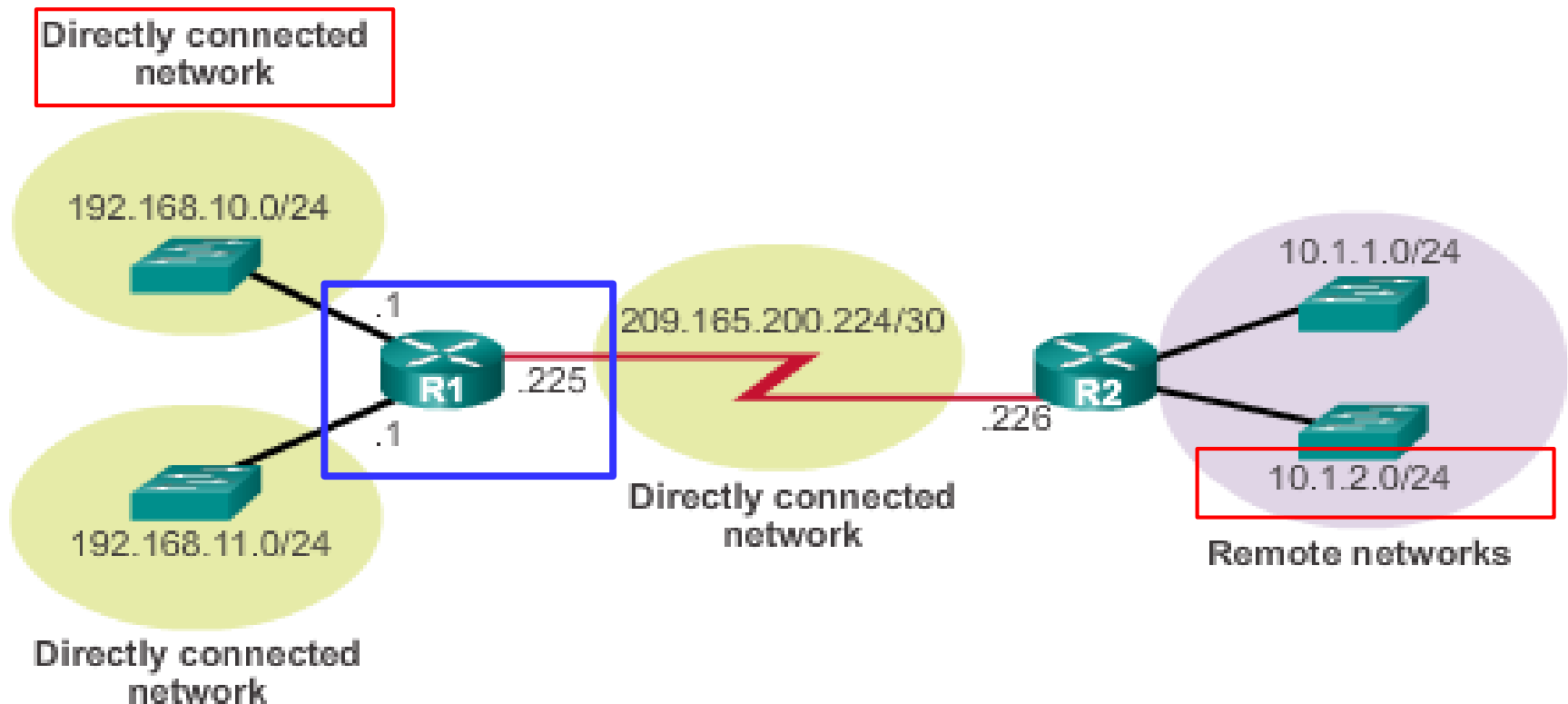
The Routing Table



The Routing Table – R1

A routing **table** is a file stored in **RAM** that contains information about:

- **R1** **Directly connected** routes
- **R1** **Remote** routes
- **Network** or **next hop** associations



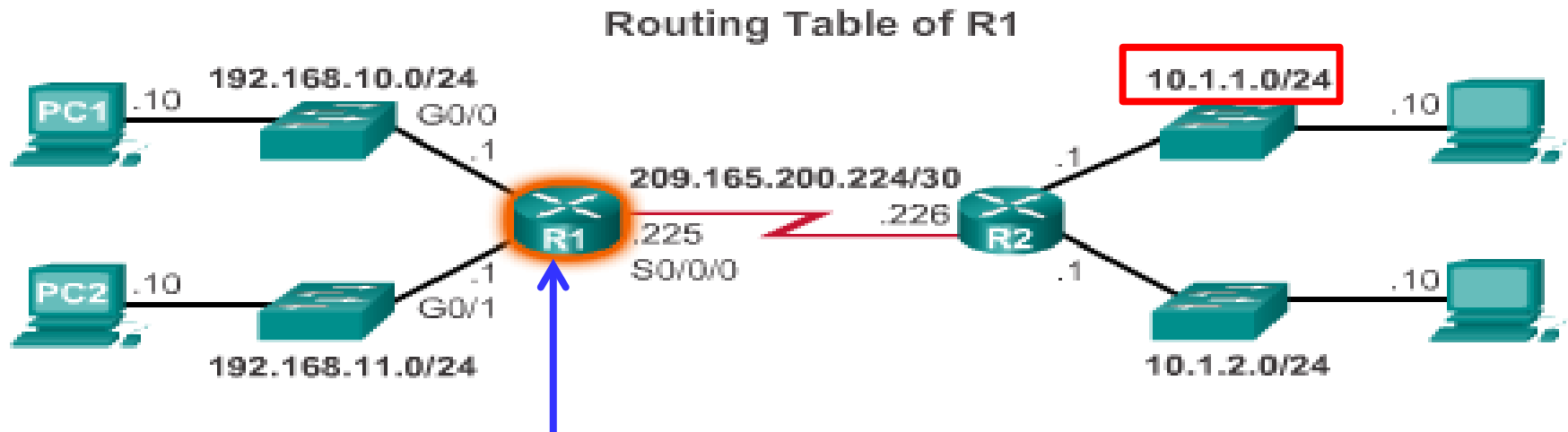
Routing Table Entries – show ip route



- **Link local route interfaces** - Added to the routing table when an interface is configured. Shows IP address of the Interface.
- **Connected interfaces** - Added to the routing table when an interface is configured and active.
- **Static routes** - Added when a route is manually configured and the exit interface is active.
- **Dynamic routes** - Added when RIP, EIGRP or OSPF are implemented and networks are identified.



Routing Table Entries – R1



```
R1#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
```

```
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
```

```
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
```

```
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
```

```
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia -
```

```
IS-IS inter area
```

```
* - candidate default, U - per-user static route, o - ODR
```

```
P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
```

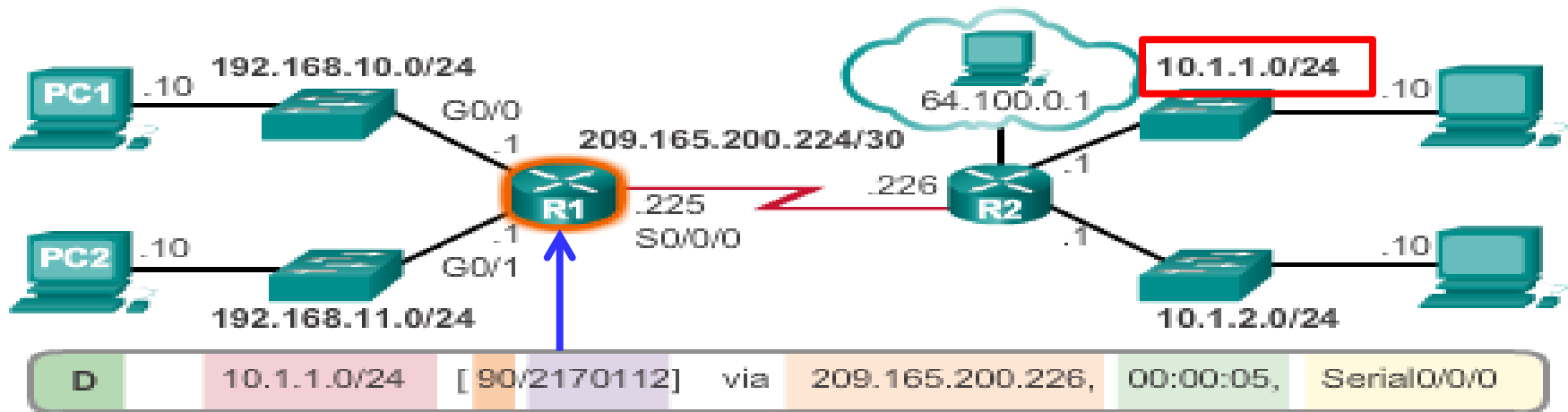
```
D      10.1.1.0/24 [90/2170112] via 209.165.200.226, 00:00:05,
```



Routing Table Entries – R1: Remote Network

Interpreting the entries in the routing table.

Remote Network Entry Identifiers



Legend

- Identifies how the network was learned by the router. **D - EIGRP**
- Identifies the destination network.
- Identifies the administrative distance (trustworthiness) of the route source.
- Identifies the metric to reach the remote network.
- Identifies the next-hop IP address to reach the remote network.
- Identifies the amount of elapsed time since the network was discovered.
- Identifies the outgoing interface on the router to reach the destination network.

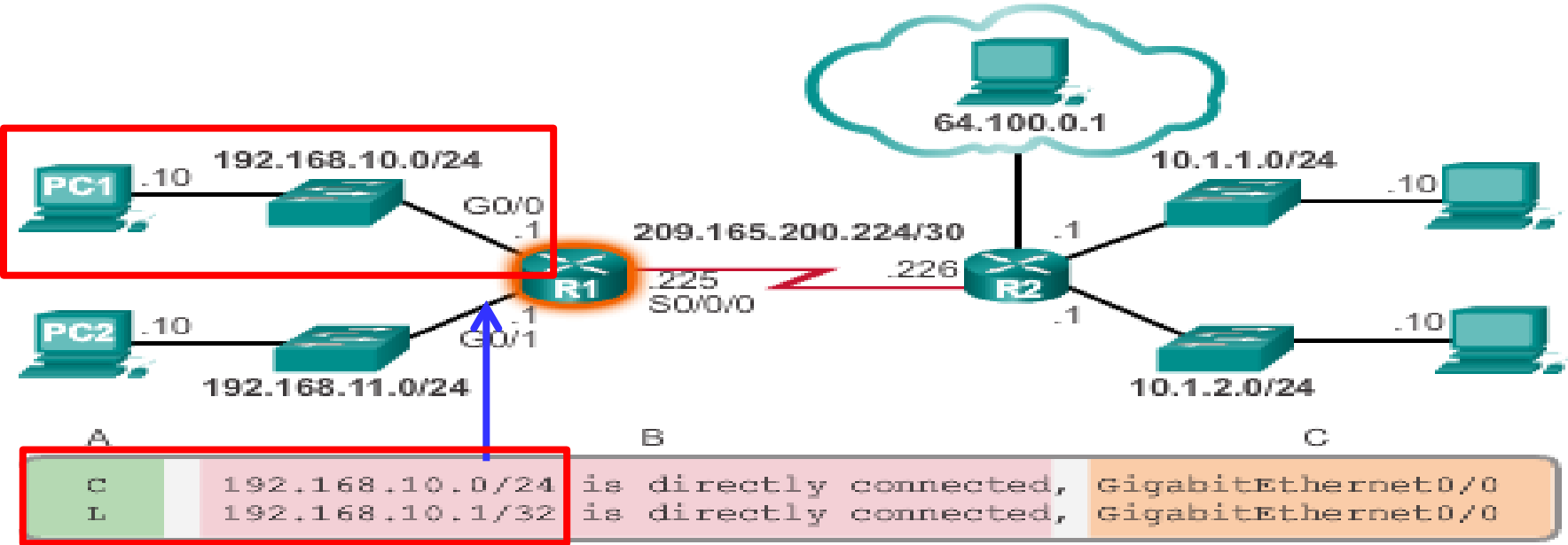
Directly Connected Routes – R1



An **active**, configured, directly connected interface creates **two routing table entries**:

- **Link Local (L)** and **Connected (C)**

Directly Connected Network Entry Identifiers



Legend

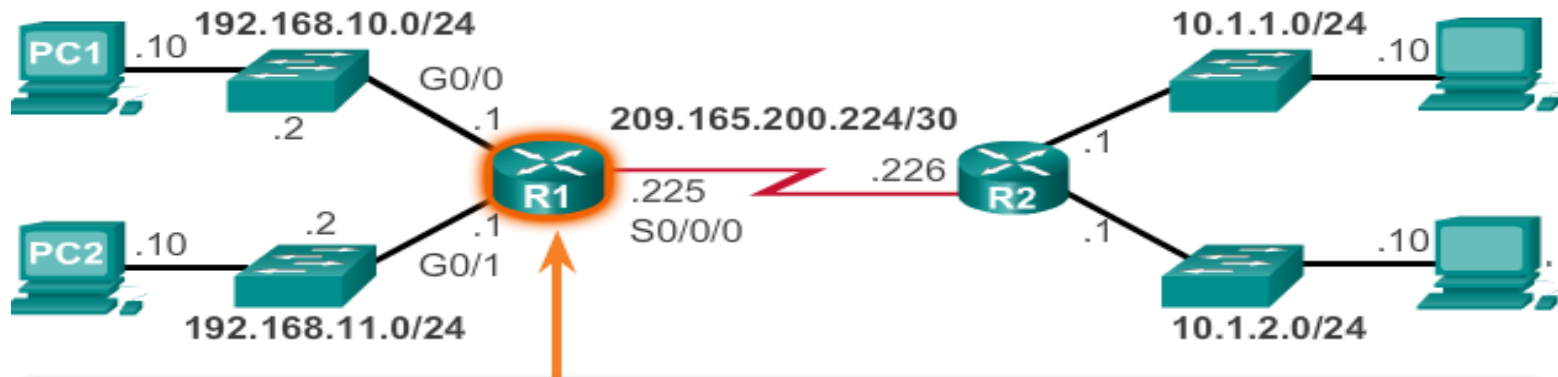
- Identifies how the network was learned by the router.
- Identifies the destination network and how it is connected.
- Identifies the interface on the router connected to the destination network.



Directly Connected – R1 Example

A routing table with the directly connected interfaces of R1 configured and activated.

Verifying the Directly Connected Routing Table Entries



```
R1# show ip route | begin Gateway
Gateway of last resort is not set

    192.168.10.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.10.0/24 is directly connected,
GigabitEthernet0/0
L       192.168.10.1/32 is directly connected,
GigabitEthernet0/0
    192.168.11.0/24 is variably subnetted, 2 subnets, 2
masks
C       192.168.11.0/24 is directly connected,
GigabitEthernet0/1
L       192.168.11.1/32 is directly connected,
GigabitEthernet0/1
```



The Static Route



- Static routes are **manually** configured
- They define an **explicit path** between two networking devices.
- Static routes must be **manually updated** if the topology changes.
- Their benefits include:
 - improved **security** – why ?
 - reduced use of **CPU** and **Memory** resources – why ?



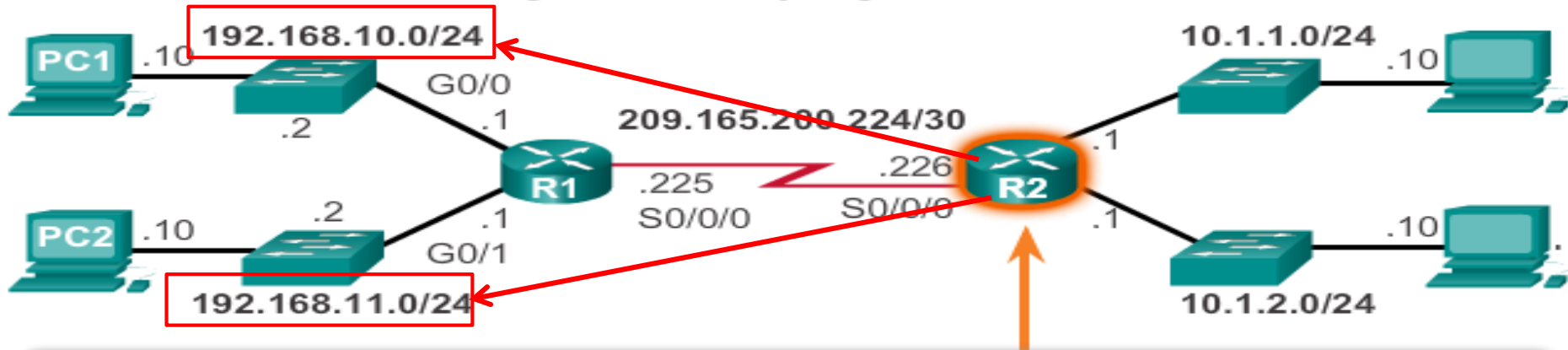
- A static route to a **specific** network:

ip route *network* *mask* {*exit-interface* | *next-hop-ip-address*}



Specific Static Routes – R2 Example

Entering and Verifying a Static Routes



```
R2(config)# ip route 192.168.10.0 255.255.255.0 s0/0/0
R2(config)# ip route 192.168.11.0 255.255.255.0 209.165.200.225
R2(config)# exit
R2#
R2# show ip route | begin Gateway
Gateway of last resort is not set

  10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C    10.1.1.0/24 is directly connected, GigabitEthernet0/0
L    10.1.1.1/32 is directly connected, GigabitEthernet0/0
C    10.1.2.0/24 is directly connected, GigabitEthernet0/1
L    10.1.2.1/32 is directly connected, GigabitEthernet0/1
S    192.168.10.0/24 is directly connected, Serial0/0/0
S    192.168.11.0/24 [1/0] via 209.165.200.225
  209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C    209.165.200.224/30 is directly connected, Serial0/0/0
```

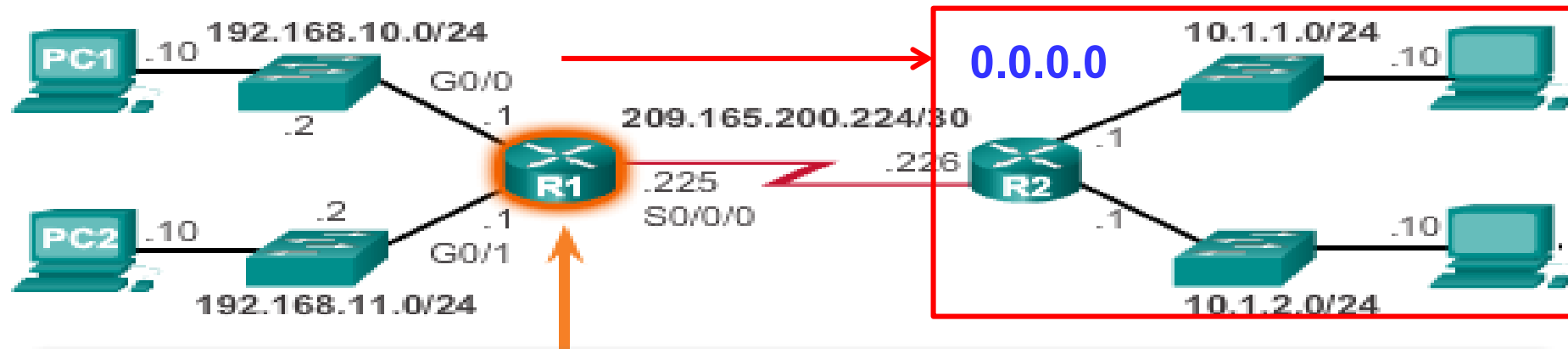


- A static **default** route
 - is used when the routing table does **not contain a path** for a destination network:
 - **ip route** 0.0.0.0 0.0.0.0 {*exit-interface* | *next-hop-ip-address*}
 - **The gateway of last resort**

Default Static Route – R1 Example



Entering and Verifying a Static Default Route



```
R1(config)#ip route 0.0.0.0 0.0.0.0 Serial0/0/0
R1(config)#exit
R1#
*Feb  1 10:19:34.483: %SYS-5-CONFIG_I: Configured from console
by console
```

```
R1#show ip route | begin Gateway
Gateway of last resort is 0.0.0.0 to network 0.0.0.0
```

```
S* 0.0.0.0/0 is directly connected, Serial0/0/0
  192.168.10.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0
  192.168.11.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.11.0/24 is directly connected, GigabitEthernet0/1
L    192.168.11.1/32 is directly connected, GigabitEthernet0/1
```



A router can learn about remote networks manually:

- Remote networks are manually entered into the routing table using static routes.

Static Routing – Used When



- it is **undesirable** to have **dynamic routing** updates forwarded across **low bandwidth** links
- for **security** reasons the **administrator needs total control** over the routes used by the router
- a **backup** to a dynamically recognized route is **necessary**

- SWINBURNE
* NE *
- CENTRE FOR
ADVANCED
INTERNET
ARCHITECTURES



Static Routing - Disadvantages

- **Initial** configuration and maintenance is **time-consuming**.
- **Configuration is error-prone**, especially in large networks.
- **Administrator intervention is required** to maintain route information.
- **Does not scale well** with growing networks; maintenance becomes cumbersome.
- **Requires complete knowledge** of the whole network for proper implementation.



Configure Static Routes - ip route Command

ip route Command Syntax

```
Router(config)#ip route network-address subnet-mask  
{ip-address | exit-intf}
```

Parameter	Description
network-address	Destination network address of the remote network to be added to the routing table.
subnet-mask	- Subnet mask of the remote network to be added to the routing table. - The subnet mask can be modified to summarize a group of networks.
ip-address	- Commonly referred to as the next-hop router's IP address. - Typically used when connecting to a broadcast media (i.e., Ethernet). - Commonly creates a recursive lookup.
exit-intf	- Use the outgoing interface to forward packets to the destination network. - Also referred to as a directly attached static route. - Typically used when connecting in a point-to-point configuration.



Configuration Options

The **Direction** in which to forward a packet, can be identified by:

- **A Next-hop route** - the **next-hop IP address** is specified.
- **A Directly Attached static route** - the router **exit interface** is specified.



Direction - Next Hop – R1

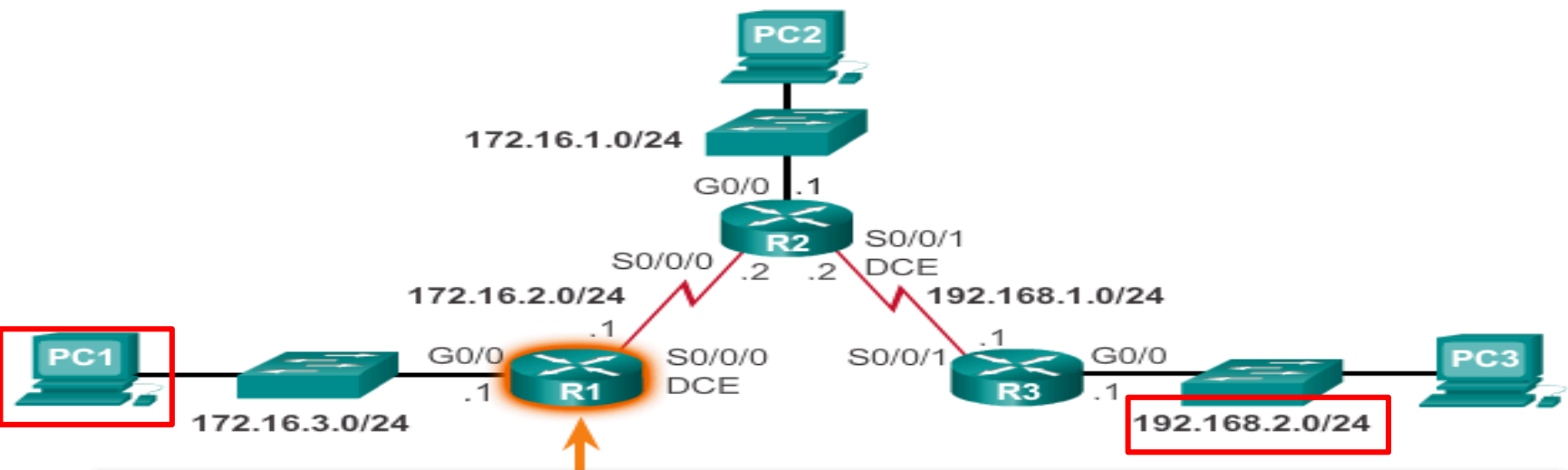
ip route 192.168.2.0 255.255.255.0 172.16.2.2
(destination) (mask) (next hop IP address)



Direction - Next Hop – R1

When a packet from PC1 is destined for the 192.168.2.0/24 network, then R1:

1. Looks for a match in the routing table and finds that it has to forward the packets to the next-hop address 172.16.2.2.
2. R1 must now determine how to reach 172.16.2.0; therefore, it searches a second time (Recursive Lookup) for a 172.16.2.0 match.



```
R1#
S 172.16.1.0/24 [1/0] via 172.16.2.2
C 172.16.2.0/24 is directly connected, Serial0/0/0
L 172.16.2.1/32 is directly connected, Serial0/0/0
C 172.16.3.0/24 is directly connected, GigabitEthernet0/0
L 172.16.3.1/32 is directly connected, GigabitEthernet0/0
S 192.168.1.0/24 [1/0] via 172.16.2.2
S 192.168.2.0/24 [1/0] via 172.16.2.2
```



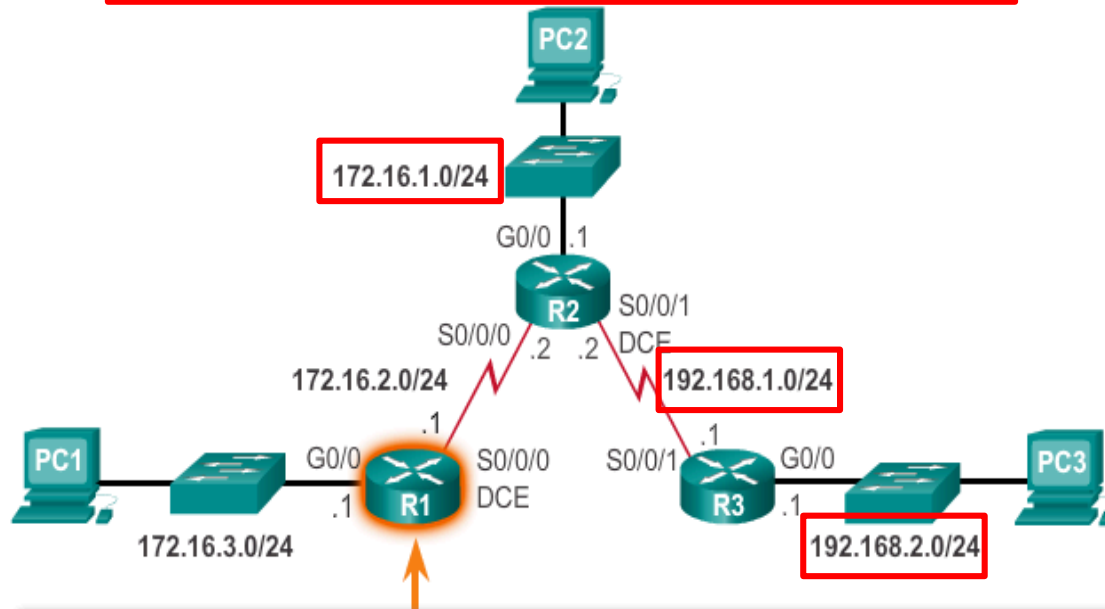
Direction - Exit Interface – R1

ip route 192.168.2.0 255.255.255.0 S0/0/0
(destination) (mask) (exit interface)



Direction - Exit Interface – R1

Configure Directly Attached Static Routes on R1



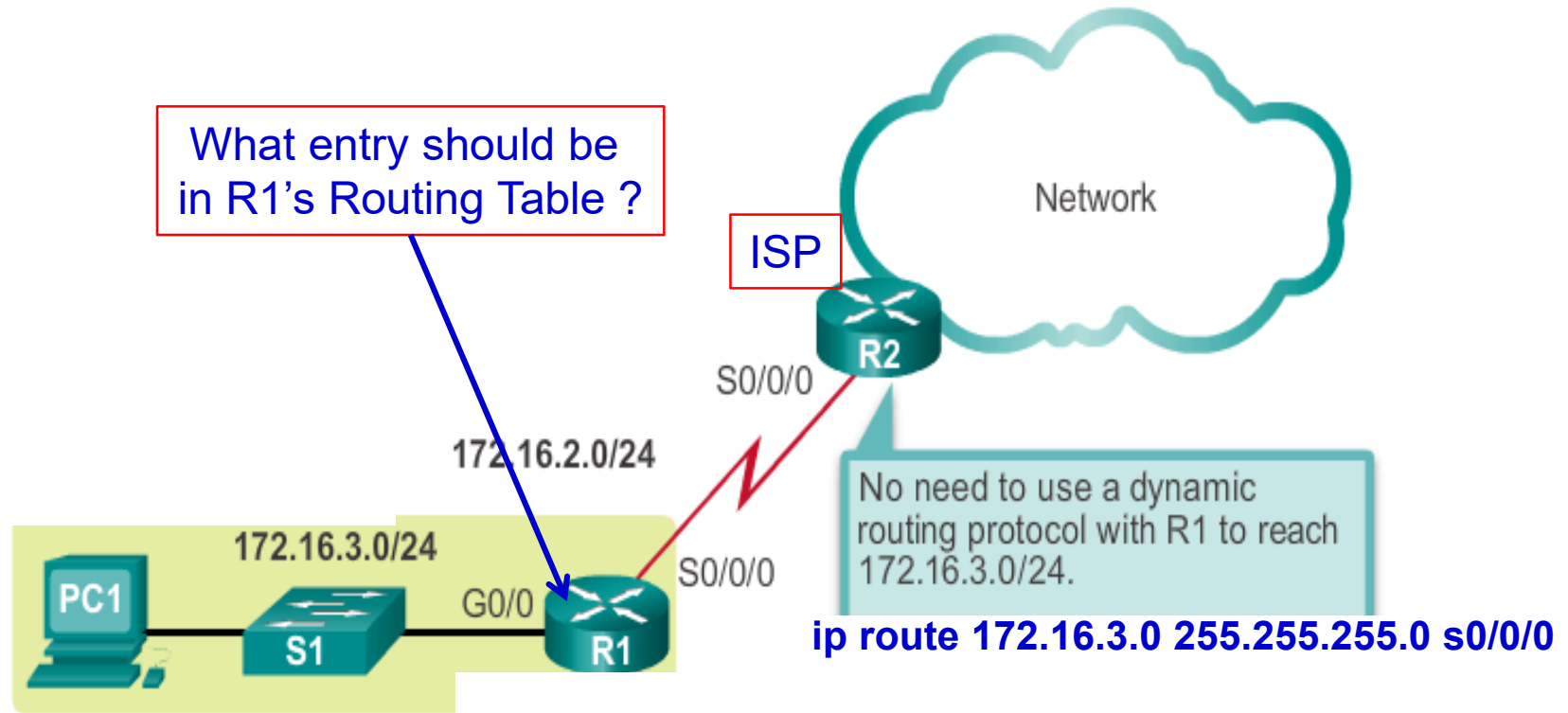
```
R1 (config) #ip route 172.16.1.0 255.255.255.0 s0/0/0
R1 (config) #ip route 192.168.1.0 255.255.255.0 s0/0/0
R1 (config) #ip route 192.168.2.0 255.255.255.0 s0/0/0
R1 (config) #
```

```
S    172.16.1.0/24 is directly connected, Serial0/0/0
C    172.16.2.0/24 is directly connected, Serial0/0/0
L    172.16.2.1/32 is directly connected, Serial0/0/0
C    172.16.3.0/24 is directly connected, GigabitEthernet0/0
L    172.16.3.1/32 is directly connected, GigabitEthernet0/0
S    192.168.1.0/24 is directly connected, Serial0/0/0
S    192.168.2.0/24 is directly connected, Serial0/0/0
R1#
```

Static Route – as a Specific route to a Network



Company Network





Static Route – as a Default Route to the world

- A **default** route is a route that matches all packets.
- A **default** route identifies the **next hop IP address or exit interface** to which the router sends all IP packets for which it does not have a **learned** or **static** route.
- A default route is simply a static route with **0.0.0.0/0** as the destination IPv4 address.
- **The Gateway of Last Resort**



Default Static Route Syntax

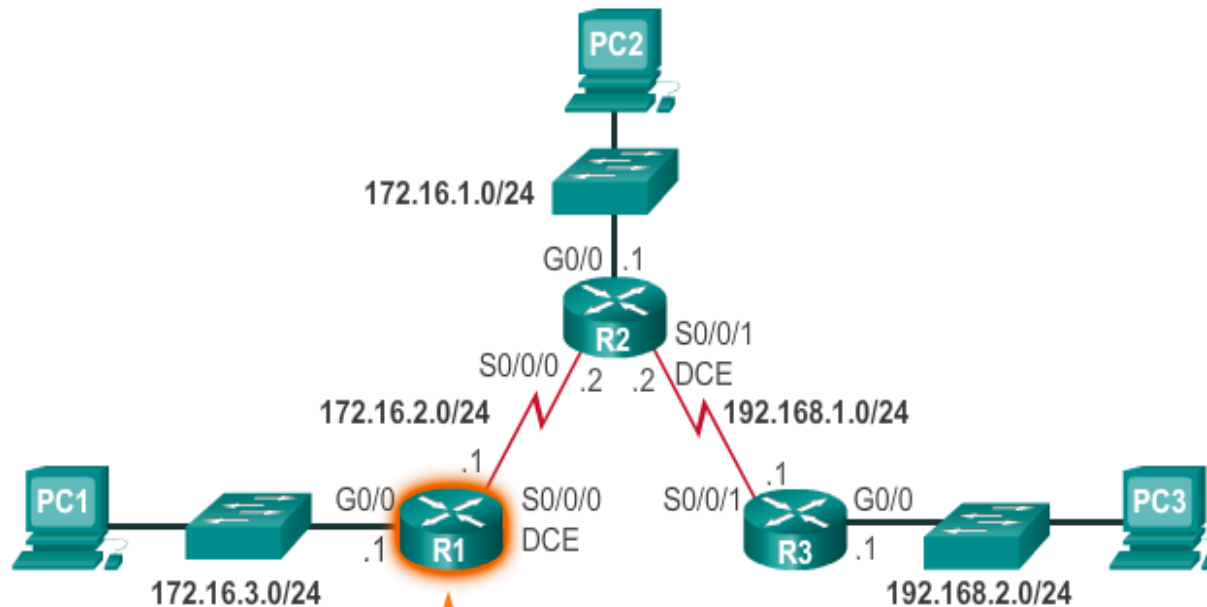
```
Router(config)#ip route 0.0.0.0 0.0.0.0 {ip-address | exit-intf}
```

Parameter	Description
0.0.0.0	Matches any network address.
0.0.0.0	Matches any subnet mask.
ip-address	• Commonly referred to as the next-hop router's IP address. • Typically used when connecting to a broadcast media (i.e., Ethernet). • Commonly creates a recursive lookup.
exit-intf	• Use the outgoing interface to forward packets to the destination network. • Also referred to as a directly attached static route. • Typically used when connecting in a point-to-point configuration.

Configure a Default Route



Configuring a Default Static Route



```
R1(config)# ip route 0.0.0.0 0.0.0.0 172.16.2.2
R1(config)#
```



CIDR – Route Summarization

- **C**lassless **I**nter-**D**omain **R**outing RIPv2, EIGRP, OSPF
- **CIDR** allows **Route summarization** with masks that are **smaller** than that of the **default class full** (**A /8**, **B /16**, **C /24**) mask.
- This type of **Route summarization** helps **reduce** the number of **entries** in **routing updates** and entries in **local routing tables**.

Static Route – as a Summary Route to multiple destinations

Route summarization, also known as,

route aggregation or supernetting

is the process of advertising

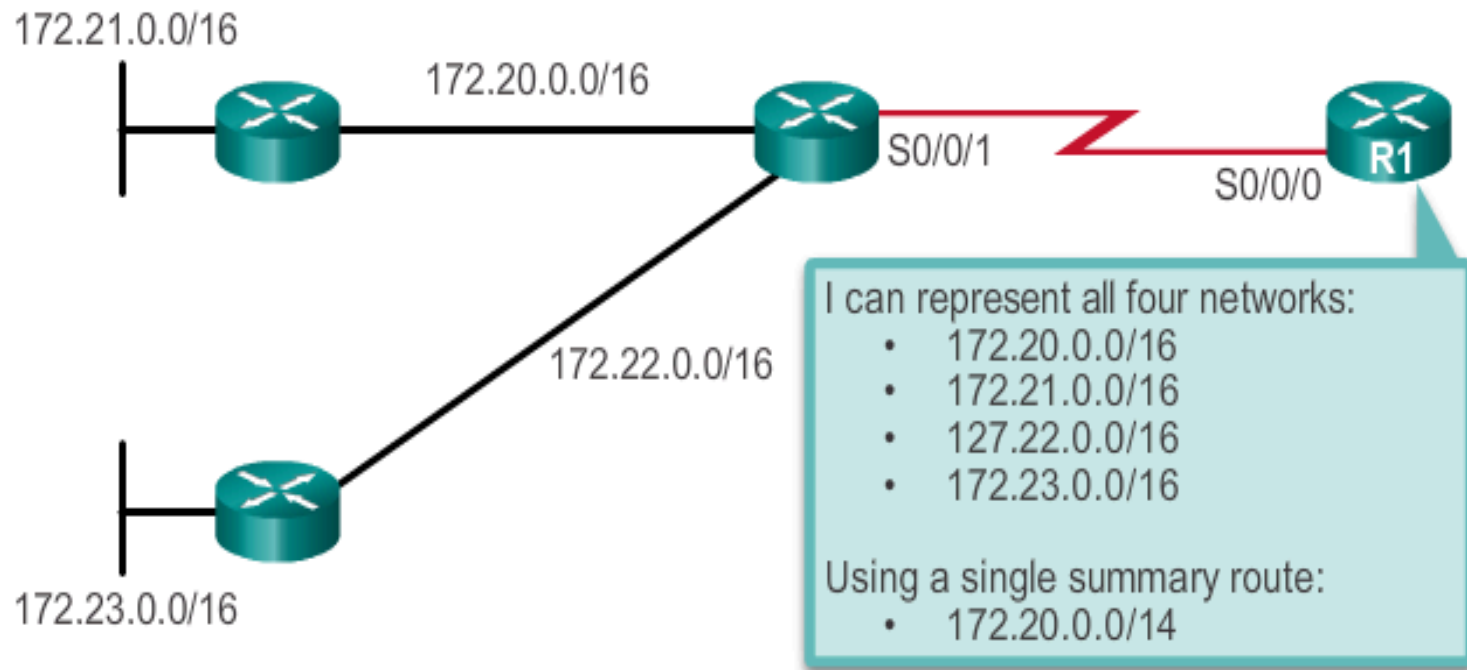
a contiguous set of addresses as a single address

with a less-specific (shorter) subnet mask



Static Route – as Summary Route

Using One Summary Static Route



The mask /14 is less than Class B default mask /16

Summary Route - Calculate a Summary Route



Calculating a Route Summary

Step 1: List networks in binary format.

172.20.0.0	10101100 . 00010100 . 00000000 . 00000000
172.21.0.0	10101100 . 00010101 . 00000000 . 00000000
172.22.0.0	10101100 . 00010110 . 00000000 . 00000000
172.23.0.0	10101100 . 00010111 . 00000000 . 00000000

Step 2: Count the number of far-left matching bits to determine the mask.

Answer: 14 matching bits = /14 or 255.252.0.0

Step 3: Copy the matching bits and add zero bits to determine the network address.

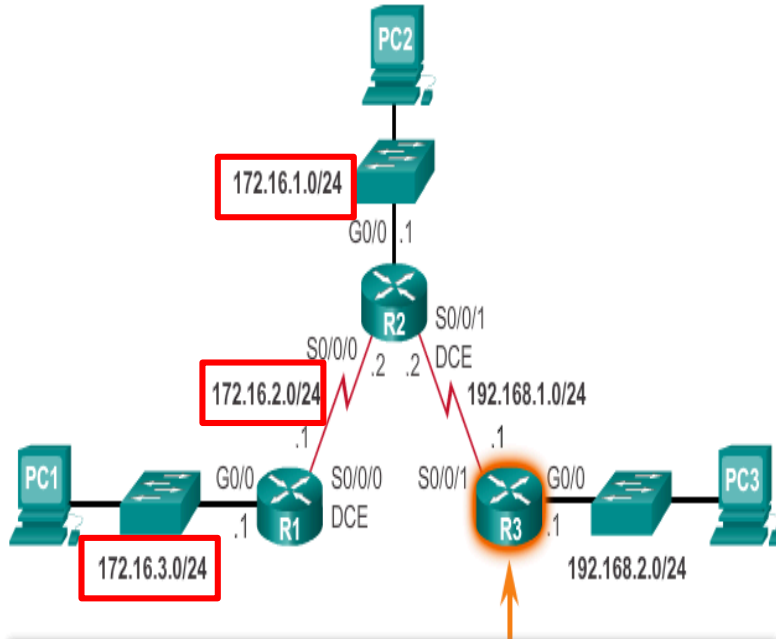
10101100 . 00010100 . 00000000 . 00000000
└── Copy ─┘ └── Add zero bits ─┘

Answer: 172.20.0.0



Summary Route - Example

Verify the Routing Table



```
R3# show ip route static | begin Gateway
```

```
Gateway of last resort is not set
```

```
172.16.0.0/24 is subnetted, 3 subnets
```

```
S    172.16.1.0 [1/0] via 192.168.1.2
```

```
S    172.16.2.0 [1/0] via 192.168.1.2
```

```
S    172.16.3.0 [1/0] via 192.168.1.2
```

Summary Boundary

Routes that can be summarized

First 22 bits are the same

Some bits are different

172.16.1.0
172.16.2.0
172.16.3.0

10101100.00010000.00000000 01.00000000
10101100.00010000.00000000 10.00000000
10101100.00010000.00000000 11.00000000

172.16.0.0
255.255.252.0

10101100.00010000.00000000 00.00000000
11111111.11111111.11111111 00.00000000

Summarized into one route

/22

172.16.0.0

255.255.252.0

Static Route – as a Backup Route



Create

a **Backup** route

in case

a **Primary** route

(link)

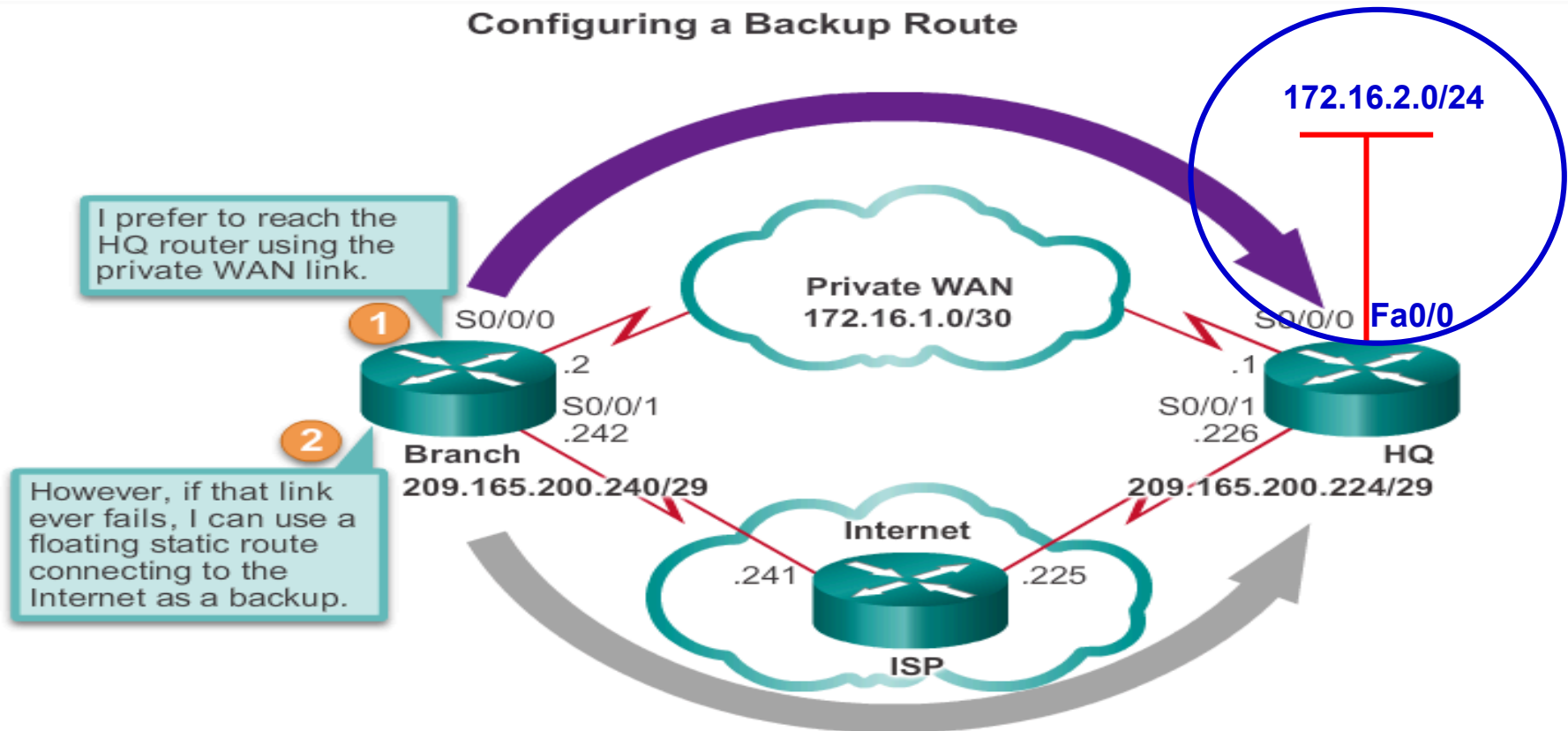
fails.



Static Route - as a Backup Route

- Used to provide a **Backup** path to a primary route, in the event of a **link failure**.
- Is only used when the primary route is not available.
- Is configured with a **higher administrative distance** than the primary route.

Configuring a Backup Route

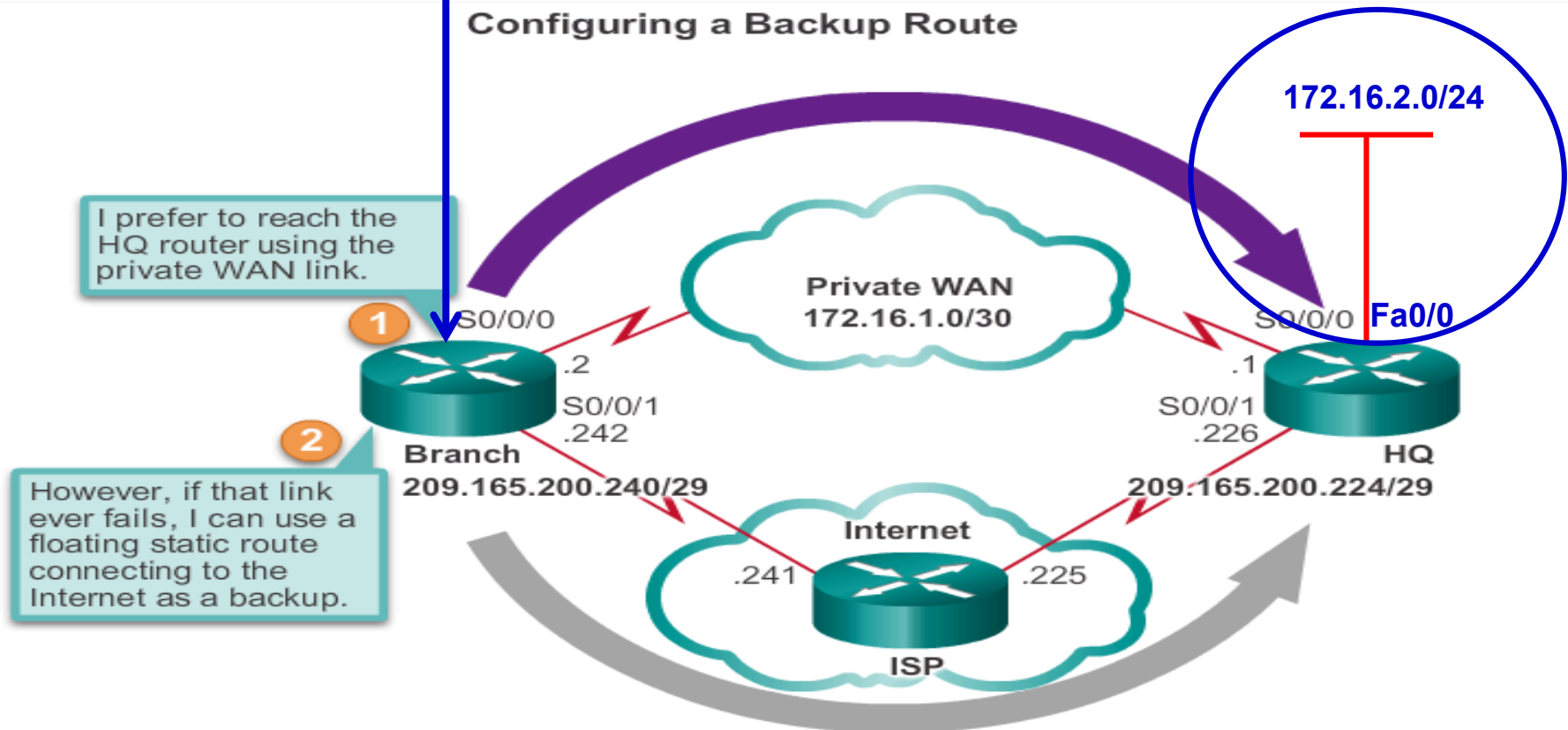




Static Route - as a Backup Route

- Branch and HQ routers use **EIGRP** with **AD 90**
- Configure Floating Static on Branch

```
ip route 172.16.2.0 255.255.255.0 S0/0/1 91
```



Verify a Static Route



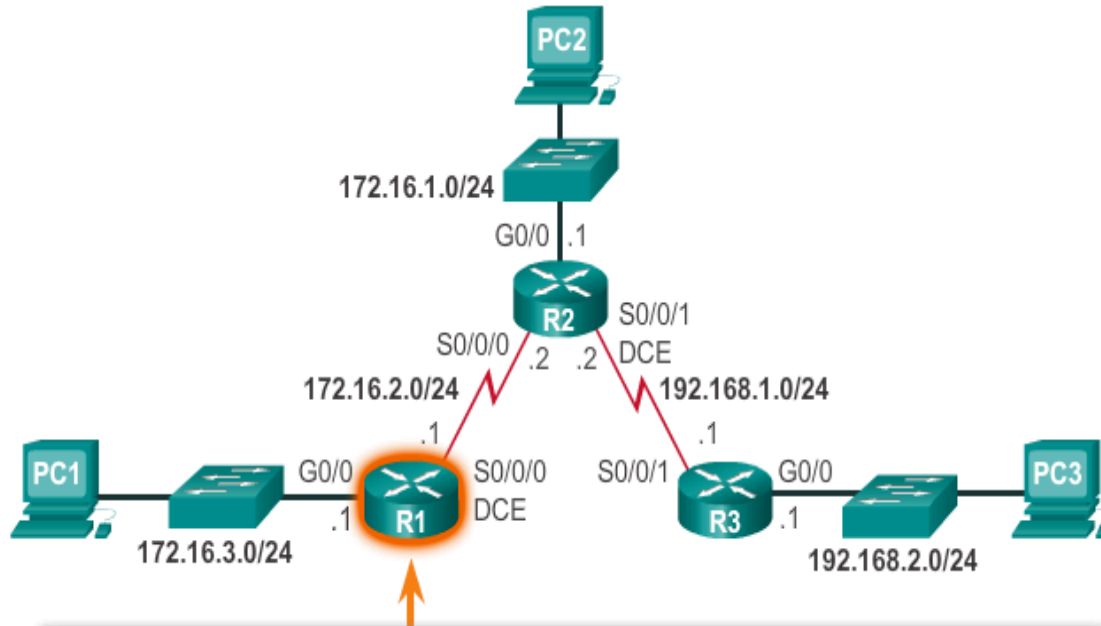
Along with **ping** and **tracert**, useful commands to verify static routes include:

- **show ip route**
- **show ip route static**
- **show ip route network**



Verify a Default Route – show ip route static

Verifying the Routing Table of R1



```
R1#show ip route static
```

Codes: L - local, C - connected, S - static, R - RIP,
M - mobile, B - BGP, D - EIGRP,
EX - EIGRP external, O - OSPF, IA -
N1 - OSPF NSSA external type 1,
N2 - OSPF NSSA external type 2,
E1 - OSPF external type 1,
E2 - OSPF external type 2, i - IS-
su - IS-IS summary, L1 - IS-IS level

* - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route,
H - NHRP, l - LISP, + - replicated route,
% - next hop override

Gateway of last resort is 172.16.2.2 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 172.16.2.2

R1#



From Routing Tables to Network Topology

Network Topology Test



- You need to **understand** the **contents** of the **routing table** to determine:
 - the **Topology** of the network
 - if there is a **Problem** with the **operation** of the **routing protocol**

Routing Table Entries – show ip route



The contents of the routing table:

- **Link local route interfaces** - Added to the routing table when an interface is configured. Shows IP address of the Interface.
- **Connected interfaces – directly connected networks** Added to the routing table when an interface is configured and active.
- **Static routes** - Added when a route is manually configured and the exit interface is active.
- **Dynamic routing protocol – remote networks** Added when RIP, EIGRP or OSPF are implemented and networks are identified.

Given Routing Tables for R1, R2 and R3

1. Draw a Network Topology Diagram

- Show all interfaces with their ip addresses
- Show all LAN subnets with their subnet addresses
- Show all Serial subnets with their subnet addresses

2. Answer Questions on the information contained in the routing tables



R1 Router - Routing Table

R1# show ip route

10.0.0.0/8 is **variably subnetted**, 3 subnets, 2 masks

C 10.1.1.0/30 is directly connected, Serial0/0/0

L 10.1.1.1/32 is directly connected, Serial0/0/0

R 10.2.2.0/30 [120/1] via 10.1.1.2, 00:00:21, Serial0/0/0

172.30.0.0/16 is **variably subnetted**, 2 subnets, 2 masks

C 172.30.10.0/24 is directly connected, FastEthernet0/1

L 172.30.10.1/32 is directly connected, FastEthernet0/1

R2 Router - Routing Table

R2# show ip route

10.0.0.0/8 is **variably subnetted**, 4 subnets, 2 masks

C 10.1.1.0/30 is directly connected, Serial0/0/0

L 10.1.1.2/32 is directly connected, Serial0/0/0

C 10.2.2.0/30 is directly connected, Serial0/0/1

L 10.2.2.2/32 is directly connected, Serial0/0/1

```
R 172.30.0.0/16 [120/1] via 10.2.2.1, 00:00:23, Serial0/0/1
    [120/1] via 10.1.1.1, 00:00:09, Serial0/0/0
```

209.165.201.0/24 is **variably subnetted**, 2 subnets, 2 masks

C 209.165.201.0/24 is directly connected, FastEthernet0/0

L 209.165.201.1/32 is directly connected, FastEthernet0/0



```

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks
C    10.2.2.0/30 is directly connected, Serial0/0/1
L    10.2.2.1/32 is directly connected, Serial0/0/1
R    10.1.1.0/30 [120/1] via 10.2.2.2, 00:00:23, Serial0/0/1

172.30.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.30.30.0/24 is directly connected, FastEthernet0/1
L    172.30.30.1/32 is directly connected, FastEthernet0/1

```

Questions



- What does **variably subnetted** mean ?
- Given, from R2 routing table:

R 172.30.0.0/16 [120/1] via 10.2.2.1, 00:00:23, Serial0/0/1
 [120/1] via 10.1.1.1, 00:00:09, Serial0/0/0

- What does [120/1] tell Us ?
 - What does 00:00:23 tell Us ?
 - Why are RIP updates from 172.30.0.0/16 coming from two different interfaces ?
-
- R1 cannot see 172.30.30.0/24 subnet, Why ?
 - R1 and R3 cannot see network 209.165.201.0/24, Why ?



THE END